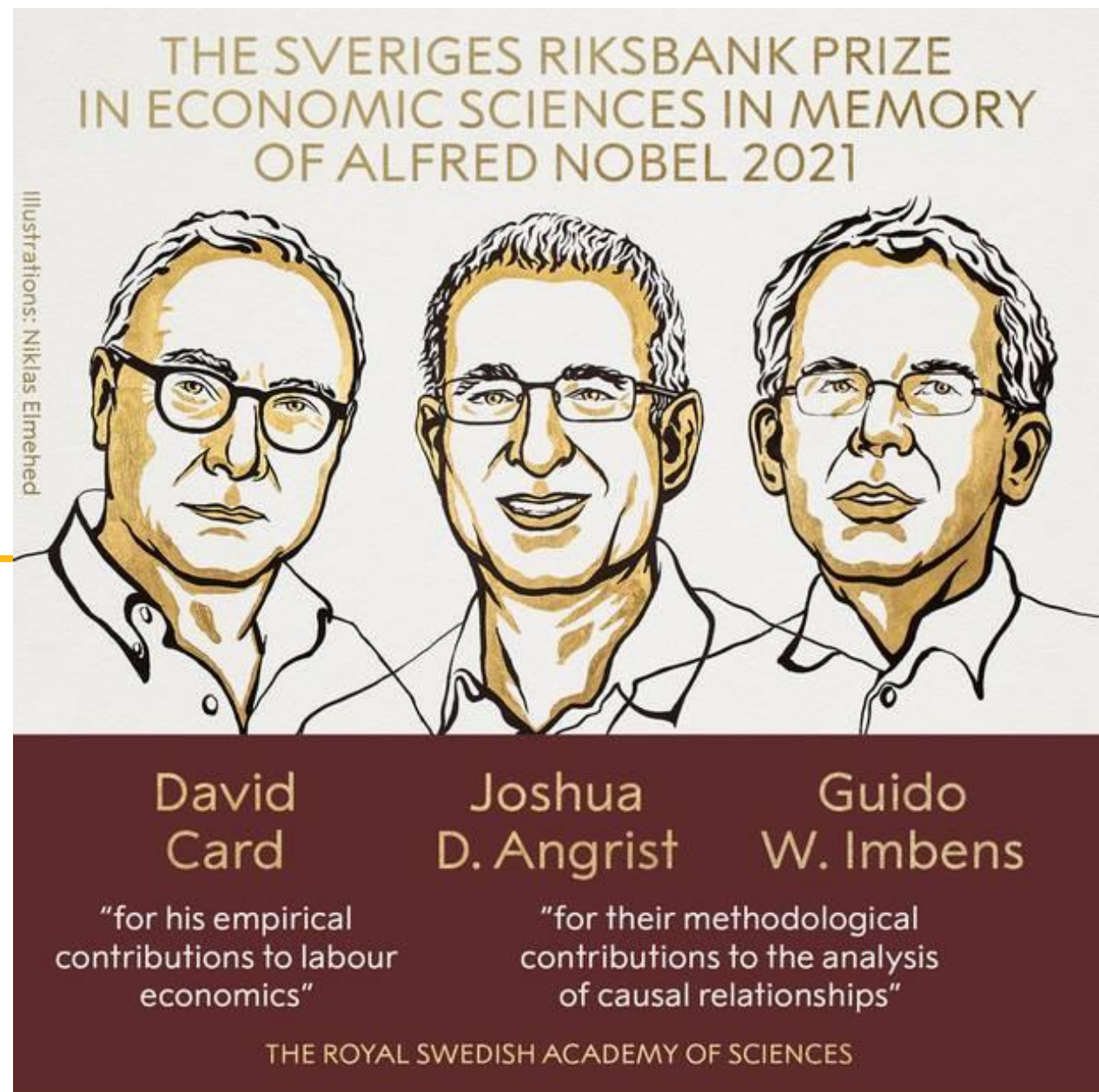


2021 노벨경제학상 심층분석 : 그들은 인과추론을 어떻게 혁신하였는가?





Scientific Background on the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2021

ANSWERING CAUSAL QUESTIONS USING OBSERVATIONAL DATA

The Committee for the Prize in Economic Sciences in Memory of Alfred Nobel

THE ROYAL SWEDISH ACADEMY OF SCIENCES, founded in 1739, is an independent organisation whose overall objective is to promote the sciences and strengthen their influence in society. The Academy takes special responsibility for the natural sciences and mathematics, but endeavours to promote the exchange of ideas between various disciplines.

BOX 50005 LILLA FRESCATIVÄGEN 4 AI, SE-104 05 STOCKHOLM, SWEDEN
TEL +46 8 673 95 00, KVA@KVA.SE • WWW.KVA.SE

ANSWERING CAUSAL QUESTIONS USING OBSERVATIONAL DATA

Most applied science is concerned with uncovering causal relationships. In many fields, randomized controlled trials (RCT's) are considered the gold standard for achieving this. The systematic use of RCT's to study causal relationships — assessing the efficacy of a medical treatment for example — has resulted in tremendous welfare gains in society. However, due to financial, ethical, or practical constraints, many important questions — particularly in the social sciences — cannot be studied using a controlled randomized experiment. For example, what is the impact of school closures on student learning and the spread of the COVID-19 virus? What is the impact of low-skilled immigration on employment and wages? How do institutions affect economic development? How does the imposition of a minimum wage affect employment? In answering these types of questions, researchers must rely on observational data, i.e., data generated without controlled experimental variation. But with observational data, a fundamental identification problem arises: the underlying cause of any correlation remains unclear. If we observe that minimum wages and unemployment correlate, is this because a minimum wage causes unemployment? Or because unemployment and lower wage growth at the bottom of the wage distribution leads to the introduction of a minimum wage? Or because of a myriad of other factors that affect both unemployment and the decision to introduce a minimum wage? Moreover, in many settings, randomized variation by itself is not sufficient for identification of an average treatment effect.

This year's Prize in Economic Sciences rewards three scholars: David Card of the University of California, Berkeley, Joshua Angrist of Massachusetts Institute of Technology, and Guido Imbens of Stanford University. The Laureates' contributions are separate but complementary. Starting with a series of paper from the early 1990s, David Card began to analyze a number of core questions in labor economics using “natural experiments”, i.e., a study design in which the units of analysis are exposed to as good as random variation caused by nature, institutions, or policy changes. These initial studies — on the minimum wage, on the impact of immigration, and on education policy — challenged conventional wisdom, and were also the starting point of an iterative process of replications, new empirical studies, and theoretical work, with Card remaining a core contributor. Thanks to this work, we have gained a much deeper understanding of how labor markets operate.

In the mid-1990s, Joshua Angrist and Guido Imbens made fundamental contributions to the challenge of estimating an average treatment effect. In particular, they analyzed the realistic scenario when individuals are affected differently by the treatment and choose whether to comply with the assignment generated by the natural experiment. Angrist and Imbens showed that even in this general setting it is possible to estimate a well-defined treatment effect — the local average treatment effect (LATE) — under a set of minimal (and in many cases empirically plausible) conditions. In deriving their key results, they merged the instrumental variables (IV) framework, common in economics, with the potential-outcomes framework for causal inference, common in statistics. Within this framework, they clarified the core identifying assumptions in a causal design and provided a transparent way of investigating the sensitivity to violations of these assumptions.

The combined contribution of the Laureates, however, is larger than the sum of the individual parts. Card's studies from the early 1990s showcased the power of exploiting natural experiments to uncover causal effects in important domains. This early work thus played a crucial role in shifting the focus in empirical research using observational data towards relying on quasi-experimental variation to establish causal effects. The framework developed by Angrist and Imbens, in turn, significantly altered how researchers approach empirical questions using data generated from either natural experiments or randomized experiments with incomplete compliance to the assigned treatment. At the core, the LATE interpretation clarifies what can and

노벨상 선정 이유에 관한
46페이지 분량의 보고서

<https://www.nobelprize.org/uploads/2021/10/advanced-economicsciencesprize2021.pdf>

ANSWERING CAUSAL QUESTIONS USING OBSERVATIONAL DATA

Most applied science is concerned with uncovering causal relationships. In many fields, randomized controlled trials (RCT's) are considered the gold standard for achieving this. The systematic use of RCT's to study causal relationships — assessing the efficacy of a medical treatment for example — has resulted in tremendous welfare gains in society. However, due to financial, ethical, or practical constraints, many important questions — particularly in the social sciences — cannot be studied using a controlled randomized experiment. For example, what is the impact of school closures on student learning and the spread of the COVID-19 virus? What is the



“Taken together, therefore, the Laureates’ contributions have played a central role in establishing the so-called **design-based approach** in economics. This approach — aimed at **emulating a randomized experiment to answer a causal question using observational data** — has transformed applied work and improved researchers’ ability to answer causal questions of great importance for economic and social policy using observational data.” (P. 2)

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<https://www.nobelprize.org/uploads/2021/10/advanced-economicsciencesprize2021.pdf>

키워드 1. 자연 실험 (Natural Experiment)



인과추론의 데이터과학

Nobel Prize in economics goes to 'natural experiments' pioneers

US economists David Card, Joshua Angrist and Guido Imbens win the 2021 Nobel Prize in economics.



Natural experiments use real-life situations to work out effects on the world [Claudio Bresciani/TT News Agency/via Reuters]

인과추론 연구 읽기 시리즈

<https://www.aljazeera.com/news/2021/10/11/nobel-prize-in-economics-goes-to-natural-experiments-pioneers>

NEWS | 13 October 2021

Nobel-winning 'natural experiments' approach made economics more robust

nature

Joshua Angrist, Guido Imbens and David Card share the prize for finding a way to identify cause and effect in social science.

Philip Ball



<https://www.nature.com/articles/d41586-021-02799-7>

키워드 2. 신뢰성 혁명 (Credibility Revolution)



인과추론의 데이터과학

The Economist

Free exchange

The Nobel prize in economics celebrates an empirical revolution

David Card shares this year's award with Joshua Angrist and Guido Imbens



Otto Dettmer

SOCIAL SCIENCE SPACE

2021 Nobels in Economics a Victory in the Credibility Revolution

Published on 10/12/2021 By David A. Jaeger

The Nobel committee's decision to award its economics prize for 2021 to David Card, Josh Angrist and Guido Imbens marks the culmination of a revolution in the way economists approach the world that began more than 30 years ago. Until the 1980s, experiments were uncommon in economics. Most economists who worked on the applied side of the field relied on data from surveys (like the census) or administrative sources (like social security).

<https://www.economist.com/finance-and-economics/2021/10/12/the-nobel-prize-in-economics-celebrates-an-empirical-revolution>

<https://www.socialsciencespace.com/2021/10/2021-nobels-in-economics-a-victory-in-the-credibility-revolution>

“Taking the Con out of Econometrics”



Let's Take the Con out of Econometrics

By EDWARD E. LEAMER*

Econometricians would like to project the image of agricultural experimenters who divide a farm into a set of smaller plots of land and who select randomly the level of fertilizer to be used on each plot. If some plots are assigned a certain amount of fertilizer while others are assigned none, then the difference between the mean yield of the fertilized plots and the mean yield of the unfertilized plots is a measure of the effect of fertilizer on agricultural yields. The econometrician's humble job is only to determine if that difference is large enough to suggest a real effect of fertilizer, or is so small that it is more likely due to random variation.

This image of the applied econometrician's art is grossly misleading. I would like to suggest a more accurate one. The applied econometrician is like a farmer who notices that the yield is somewhat higher under trees where birds roost, and he uses this as evidence that bird droppings increase yields. However, when he presents this finding at the annual meeting of the American Ecological Association, another farmer in the audience objects that he used the same data but came up with the conclusion that moderate

One should not jump to the conclusion that there is necessarily a substantive difference between drawing inferences from experimental as opposed to nonexperimental data. The images I have drawn are deliberately prejudicial. First, we had the experimental scientist with hair neatly combed, wide eyes peering out of horn-rimmed glasses, a white coat, and an electronic calculator for generating the random assignment of fertilizer treatment to plots of land. This seems to contrast sharply with the nonexperimental farmer with overalls, unkempt hair, and bird droppings on his boots. Another image, drawn by Orcutt, is even more damaging: “Doing econometrics is like trying to learn the laws of electricity by playing the radio.” However, we need not now submit to the tyranny of images, as many of us have in the past.

I. Is Randomization Essential?

What is the real difference between these two settings? Randomization seems to be the answer. In the experimental setting, the fertilizer treatment is “randomly” assigned

*Professor of economics, University of California-Los Angeles. This paper was a public lecture presented at the University of Toronto, January 1982. I acknowledge partial support by NSF grant SOC78-09479.

The Credibility Revolution in Empirical Economics: How Better Research Design is Taking the Con out of Econometrics

Joshua D. Angrist and Jörn-Steffen Pischke

Just over a quarter century ago, Edward Leamer (1983) reflected on the state of empirical work in economics. He urged empirical researchers to “take the con out of econometrics” and memorably observed (p. 37): “Hardly anyone takes data analysis seriously. Or perhaps more accurately, hardly anyone takes anyone else's data analysis seriously.” Leamer was not alone; Hendry (1980), Sims (1980), and others writing at about the same time were similarly disparaging of empirical practice. Reading these commentaries as late-1980s Ph.D. students, we wondered about the prospects for a satisfying career doing applied work. Perhaps credible empirical work in economics is a pipe dream. Here we address the questions of whether the quality and the credibility of empirical work have increased since Leamer's pessimistic assessment. Our views are necessarily colored by the areas of applied microeconomics in which we are active, but we look over the fence at other areas as well.

Leamer (1983) diagnosed his contemporaries' empirical work as suffering from a distressing lack of robustness to changes in key assumptions—assumptions he called “whimsical” because one seemed as good as another. The remedy he proposed was sensitivity analysis, in which researchers show how their results vary with changes in specification or functional form. Leamer's critique had a refreshing emperor's-new-clothes earthiness that we savored on first reading and still enjoy today. But we're happy to report that Leamer's complaint that “hardly anyone takes anyone else's data analysis seriously” no longer seems justified.

■ Joshua D. Angrist is Ford Professor of Economics, Massachusetts Institute of Technology, Cambridge, Massachusetts. Jörn-Steffen Pischke is Professor of Economics, London School of Economics, London, United Kingdom. Their e-mail addresses are angrist@mit.edu and s.pischke@lse.ac.uk.

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“The econometric art as it is practiced at the computer terminal involves fitting many, perhaps thousands, of statistical models. One or several that the researcher finds pleasing are selected for reporting purposes.” (p. 36)

“Hardly anyone takes data analysis seriously. Or perhaps more accurately, hardly anyone takes anyone else’s data analysis seriously.” (p. 37)

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“But, we’re happy to report that Leamer’s complaint that “hardly anyone takes anyone else’s data analysis seriously” no longer seems justified.” (p. 3)

“A clear-eyed focus on research design is at the heart of the credibility revolution in empirical economics.” (p. 6)

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인과추론 분석, 과연 얼마나 믿을 만할까?

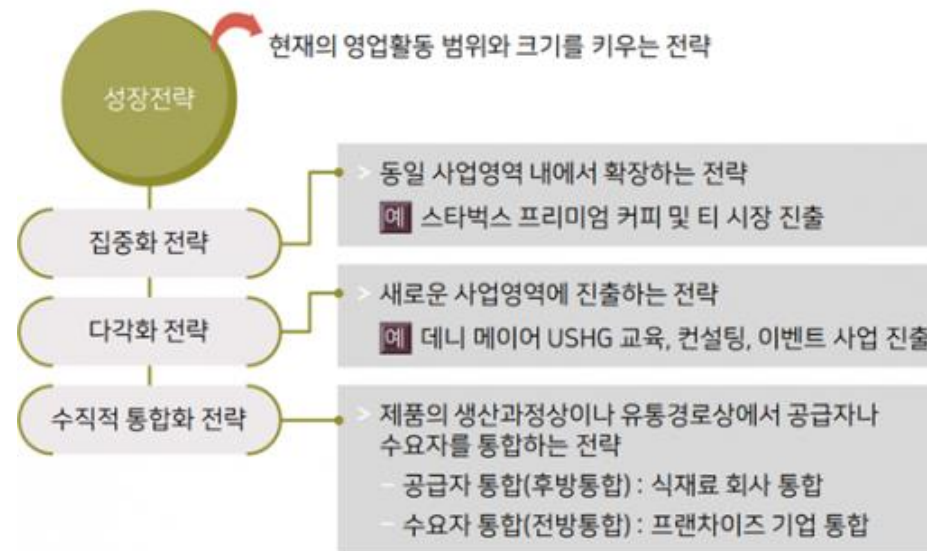


인과추론의 데이터과학

• 사례 1. 사업 다각화가 기업 가치에 미치는 영향

“Tobin's q [*measure of firm value*] and firm diversification are negatively related throughout the 1980s.” (Lang and Stulz 1994, p. 1248)

“Firms choose to diversify. We use three alternative **econometric techniques to control for the endogeneity of the diversification decision**... The diversification discount always drops, and sometimes turns into a premium.” (Campa and Kedia 2002, p. 1731)



카카오의 사업 확장

2021년

퀵서비스 카카오T 앱으로 퀵서비스 호출, 서비스 개시 2주 만에 가입 기사 10만명 돌파

꽃 배달 카카오T 앱에서 화환·꽃바구니 주문 배달

택시 기사 멤버십 월 9만9000원, 우선 배차권

여성 패션 쇼핑물 지그재그 1조원에 인수

2019년

영어 교육 육아 콘텐츠 자회사 카카오키즈, 야나두 인수

2017년

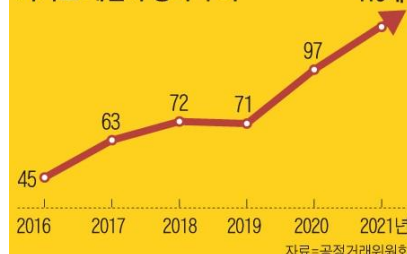
골프 카카오게임즈, 국내 스크린 골프 2위 마음골프 인수

2016년

미용실·네일샵 카카오톡 앱 내 예약

대리운전 카카오T 앱으로 대리기사 호출

카카오 계열사 증가 추이



Lang, L. H., & Stulz, R. M. 1994. Tobin's q, Corporate Diversification, and Firm Performance. *Journal of Political Economy*, 102(6), 1248-1280.
Campa, J. M., & Kedia, S. 2002. Explaining the Diversification Discount. *Journal of Finance*, 57(4), 1731-1762.

- 사례 2. 종교가 경제 성장에 미치는 영향

RELIGION AND ECONOMIC GROWTH ACROSS COUNTRIES

ROBERT J. BARRO
Harvard University

RACHEL M. MCCLEARY
Harvard University

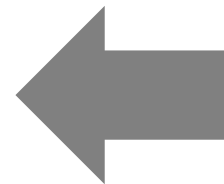
Empirical research on the determinants of economic growth typically neglects the influence of religion. To fill this gap, this study uses international survey data on religiosity for a broad panel of countries to investigate the effects of church attendance and religious beliefs on economic growth. To isolate the direction of causation from religiosity to economic performance, the estimation relies on instrumental variables suggested by an analysis in which church attendance and religious beliefs are the dependent variables. The instruments are variables for the presence of state religion and for regulation of the religion market, the composition of religious adherence, and an indicator of religious pluralism. Results show that economic growth responds positively to religious beliefs, notably beliefs in hell and heaven, but negatively to church attendance. That is, growth depends on the extent of believing relative to belonging. These results accord with a model in which religious beliefs influence individual traits that enhance economic performance. The beliefs are an output of the religion sector, and church attendance is an input to this sector. Hence, for given beliefs, higher church attendance signifies more resources used up by the religion sector.

Model Uncertainty in Sociological Research: An Application to Religion and Economic Growth

Cristobal Young
Princeton University

(당시 박사과정 학생)

Model uncertainty is pervasive in quantitative research. Classical statistical theory assumes that only one (true) model is applied to a sample of data. In practice, however,



분석 결과에
대한 반박

“Small, sensible changes in their model specification produce large changes in the results: the results are inconsistent across time, and the instrumental variables strategy suffers from a weak instrument set. Also, the observed relationship between religiosity and economic growth does not hold in the West; it is largely a feature of Asian and African countries and of countries whose data is poor quality” (p. 3)

Barro, J. Robert and Rachel M. McCleary. 2003. Religion and Economic Growth across Countries. *American Sociological Review*, 68(5), pp. 760–81.

Young, C., 2009. Model Uncertainty in Sociological Research: An Application to Religion and Economic Growth. *American Sociological Review*, 74(3), pp.380-397.

- 사례 3. ‘그냥 한 번 2백만개의 회귀분석을 돌려봤어’

I Just Ran Two Million Regressions

By XAVIER X. SALA-I-MARTIN*

Following the seminal work of Robert Barro (1991), the recent empirical literature on economic growth has identified a substantial number of variables that are partially correlated with the rate of economic growth. The basic methodology consists of running cross-sectional regressions of the form

$$(1) \quad y = \alpha + \beta_1 x_1 + \beta_2 x_2 \\ + \cdots + \beta_n x_n + \varepsilon$$

An initial answer to this question was given by Ross Levine and David Renelt (1992).¹ They applied Edward Leamer's (1985) *extreme-bounds test* to identify "robust" empirical relations in the economic growth literature. In short, the extreme-bounds test works as follows. Imagine that there is a pool of N variables that previously have been identified to be related to growth and one is interested in knowing whether variable z is "robust." One would estimate regressions of the form

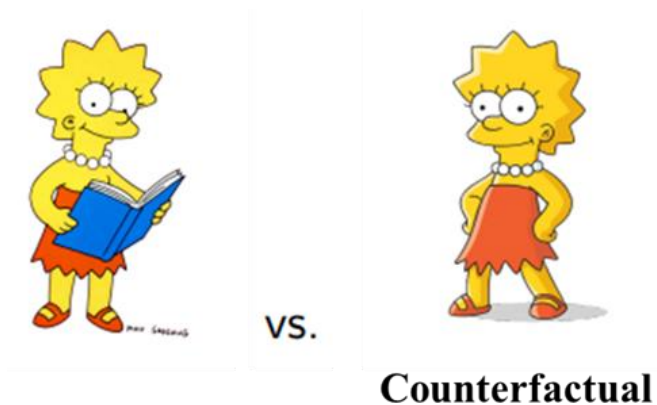
* Department of Economics, Columbia University, 420 West 118th St., New York, NY 10027, and Universitat Pompeu Fabra, Barcelona, Spain.

Martin, S.I., X.(1997). I Just Ran Two Million Regressions. *American Economic Review*, 87(2), pp.178-183.

- 기존 연구에서 경제성장에 영향을 미친다고 밝혀졌던 62개 원인 변수들 중에서 가장 중요한 3개의 통제변수를 고정한 후, 나머지 변수에 대해서 하나씩 민감성 분석을 수행함. 이 때, 나머지 58개 변수들 중에서 3개의 추가적인 통제 변수를 고려하여 총 182 만여개의 회귀분석 모델을 도출함.
- 분석 결과, 총 59개 원인 변수들 중에서 37개의 변수들의 효과가 통제변수에 따라 민감하게 달라졌다는 것을 보임.

인과추론 실증연구(데이터분석)에 대한 재현성/신뢰성 위기

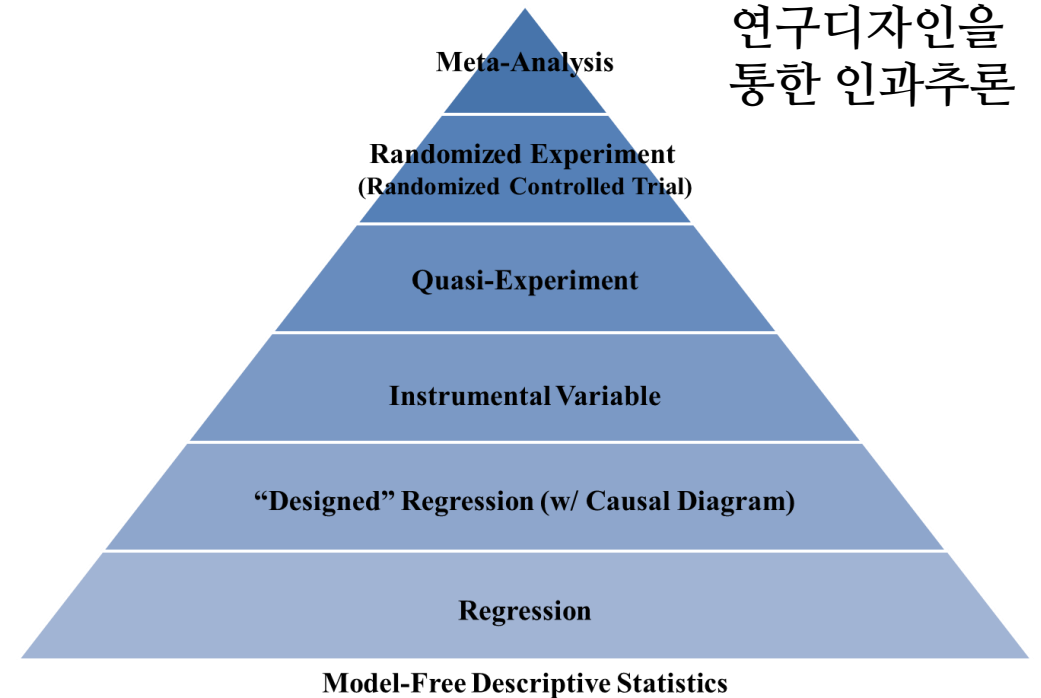
잠재적결과 프레임워크 (Rubin Causal Model)



인과 효과

= (원인이 있을 때의 결과) - (원인이 없었다면 있었을 잠재적 결과)

Level of Causal Inference



분석모형이 아닌,
연구디자인을
통한 인과추론

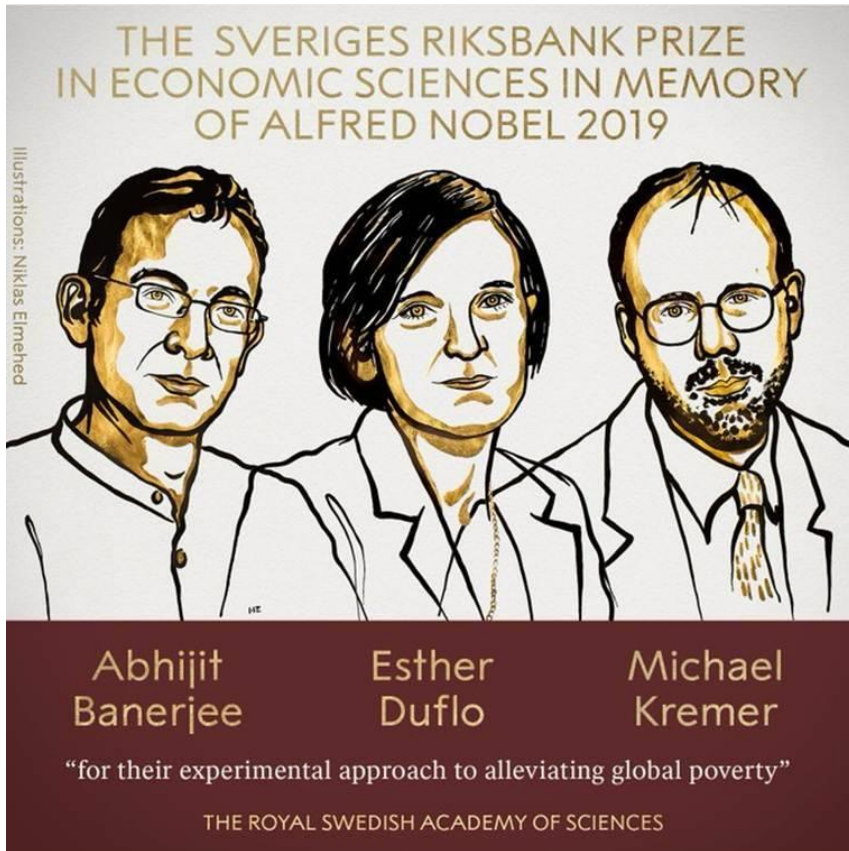
잠재적결과 프레임워크에 기반한 연구 디자인은 데이터분석 결과의 인과적 해석에 대한 신뢰성을 부여함으로써 실증 연구의 위상과 역할을 크게 향상시켰음.

인과추론을 위한 연구 디자인과 신뢰성 혁명



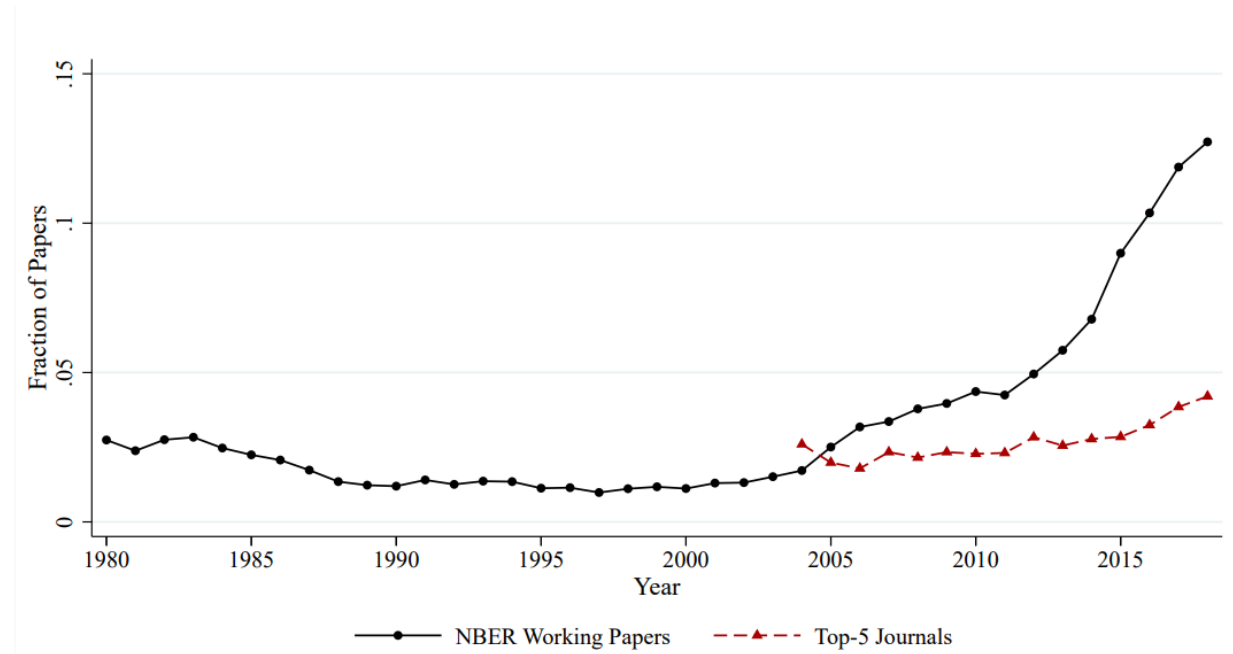
인과추론의 데이터과학

- 인과추론을 위한 가장 효과적인 연구 디자인(“Gold Standard”)은 랜덤화 실험 (Randomized Controlled Trial)
 - 2019 노벨경제학상 수상 공로가 대표적인 사례
 - 하지만, 현실의 많은 문제들에 있어 인위적인 실험이 불가능하거나, 가능하더라도 비윤리적인 경우가 많음.



경제학 논문에서 RCT 활용 비율

A: RCTs



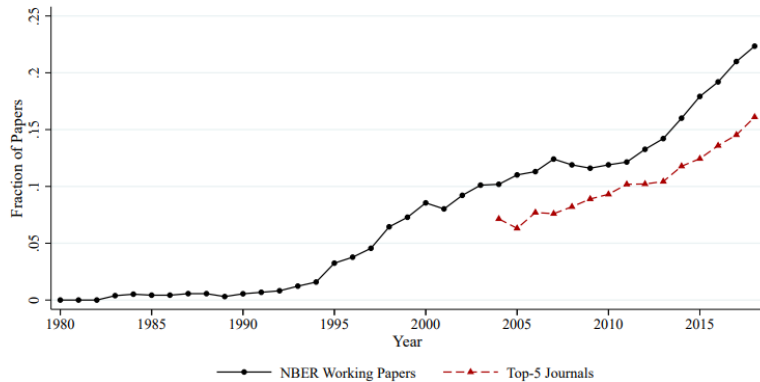
Currie, J., Kleven, H. and Zwieters, E., 2020, May. Technology and Big Data are Changing Economics: Mining Text to Track Methods. In *AEA Papers and Proceedings* (Vol. 110, pp. 42-48).

인과추론을 위한 연구 디자인과 신뢰성 혁명

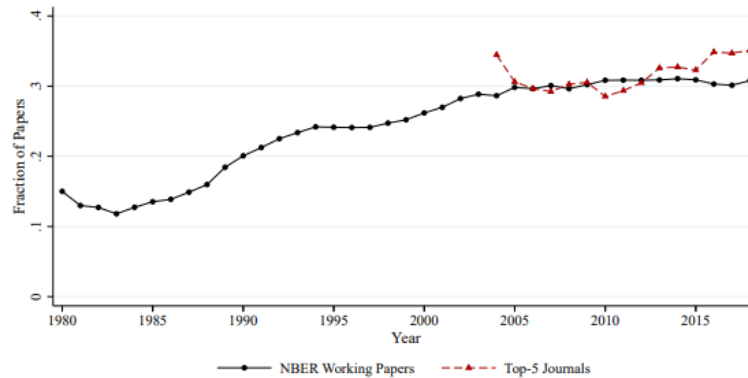
- 신뢰성 혁명의 도화선이 된 것은 실험 아닌 실험 같은 준실험 (Quasi-Experiment) 또는 자연실험 (Natural Experiment)

경제학 논문에서 준실험/자연실험 방법론 활용 비율

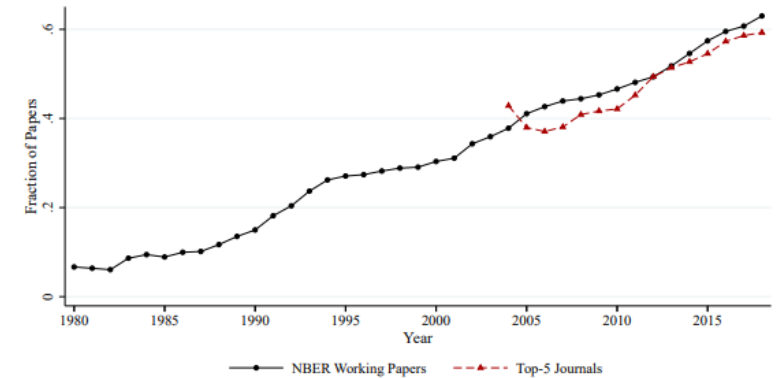
A: Difference-in-Differences



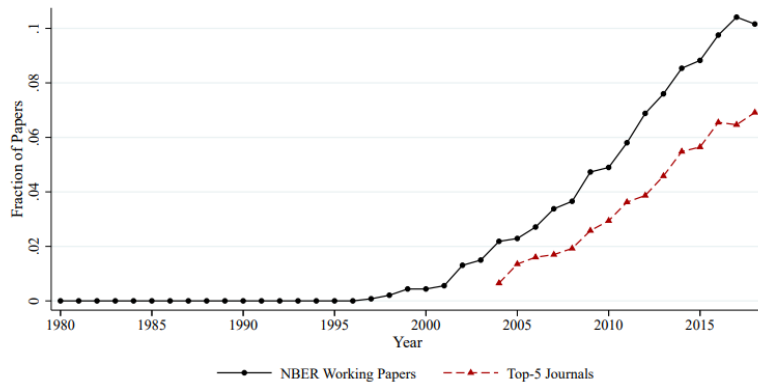
A: Instrumental Variables



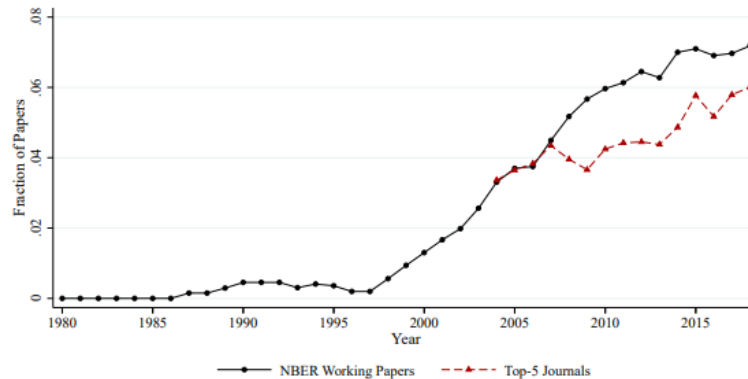
B: Fixed Effects



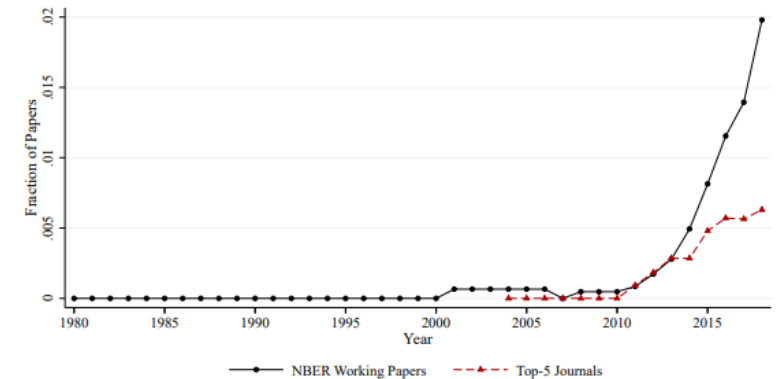
B: Regression Discontinuity



C: Matching



D: Synthetic Control



인과추론을 위한 연구 디자인과 신뢰성 혁명

- 신뢰성 혁명은 경제학 뿐 아니라, 사회과학 전분야에서 일어나고 있는 가장 큰 변화 중 하나 (실증 연구/데이터 분석)

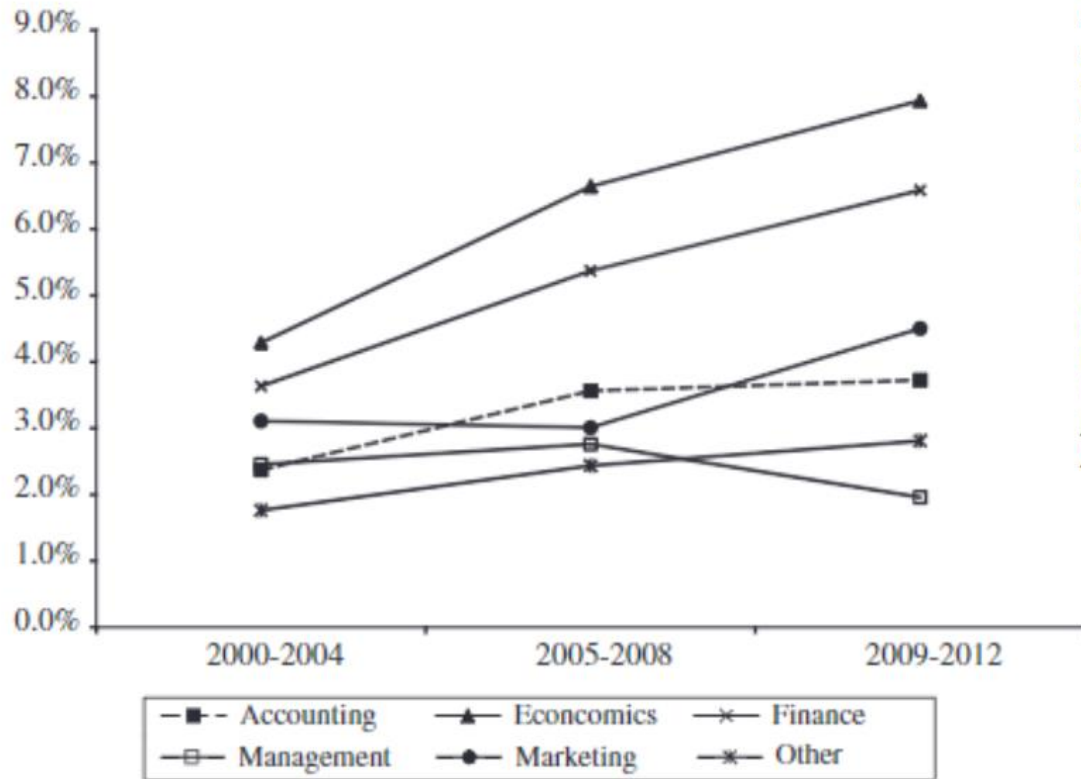
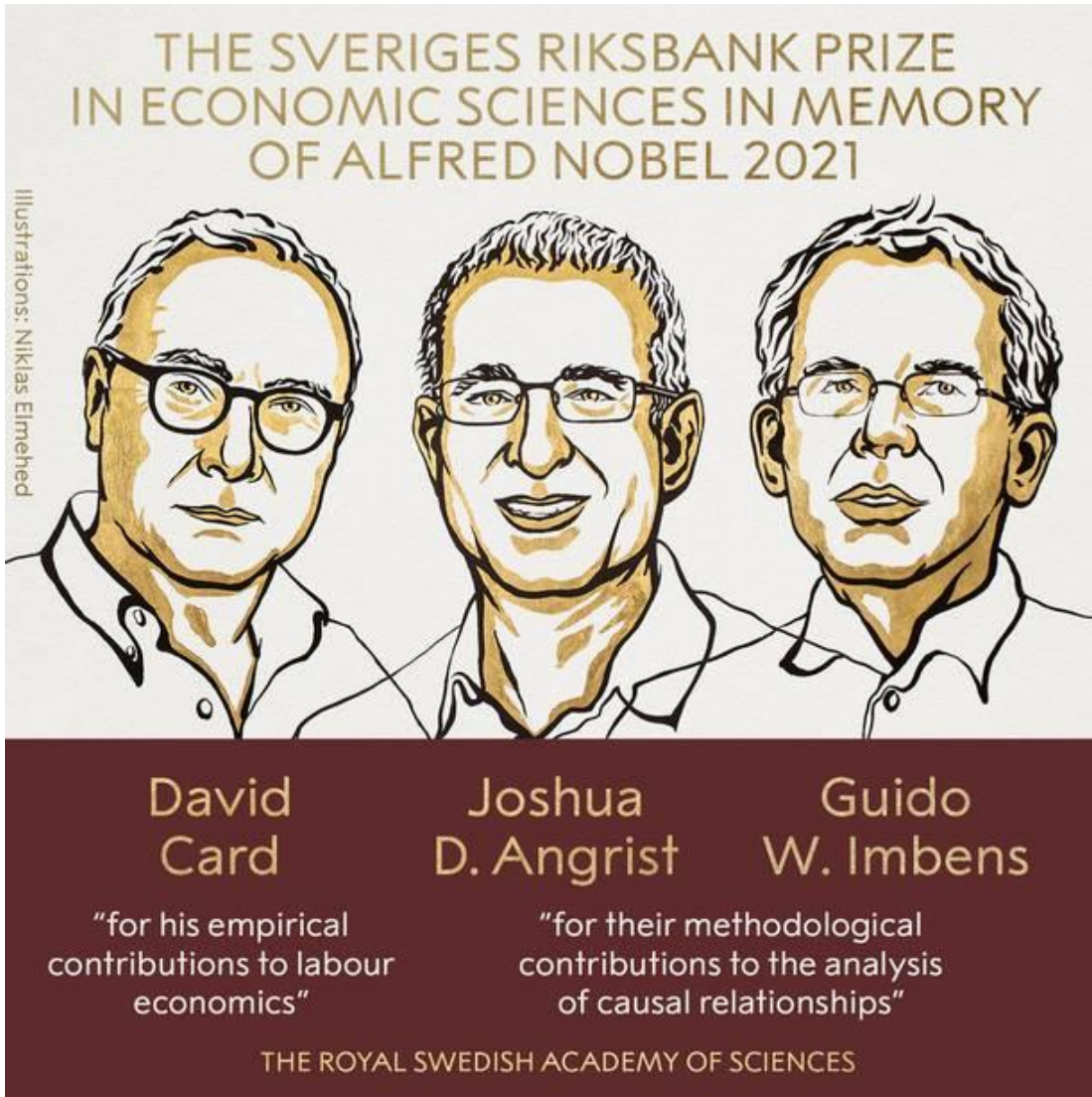


Fig. 1. Relative frequency of causal studies by field and publication period.

Journal title	Field	Journal title	Field
Academy of Management Journal	Management	Journal of Applied Psychology	Other
Academy of Management Perspectives	Management	Journal of Business Ethics	Other
Academy of Management Review	Management	Journal of Business Venturing	Other
Accounting Organizations and Society	Accounting	Journal of Consumer Psychology	Marketing
Accounting Review	Accounting	Journal of Consumer Research	Marketing
Administrative Science Quarterly	Management	Journal of Finance	Finance
American Economic Review	Economics	Journal of Financial and Quantitative Analysis	Finance
California Management Review	Management	Journal of Financial Economics	Finance
Contemporary Accounting Research	Accounting	Journal of International Business Studies	Other
Econometrica	Economics	Journal of Management Studies	Management
Entrepreneurship Theory and Practice	Other	Journal of Marketing	Marketing
Harvard Business Review	Management	Journal of Marketing Research	Marketing
Human Resource Management	Other	Journal of Operations Management	Other
Information Systems Research	Other	Journal of Political Economy	Economics
Journal of Accounting Economics	Accounting	Management Science	Other
Journal of Accounting Research	Accounting	Marketing Science	Marketing
		MIS Quarterly	Other
		Operations Research	Other
		Organization Science	Other
		Organization Studies	Other
		Organizational Behavior and Human Decision Processes	Other
		Quarterly Journal of Economics	Economics
		RAND Journal of Economics	Economics
		Review of Accounting Studies	Accounting
		Review of Financial Studies	Finance
		Strategic Management Journal	Management



2021 노벨경제학상 수상자들의 공로

- 인과추론의 개념 도입? No.
- 인과추론을 위한 새로운 방법론 개발? No.
- 계량경제학/통계학 분야에서 연구되고 활용되어 오던 인과추론 방법론들을 본격적으로 현실 문제 해결에 적용함으로써, 인과추론을 위한 연구 디자인과 데이터분석 방법론을 체계적으로 확립하는데 공헌함.
- ‘인과추론 분석의 현실 문제 적용의 지평을 넓힌 공로’

자연실험 사례 (1) 최저임금 인상이 고용에 미치는 영향 (Card and Krueger 1994)

- 뉴저지에서의 최저임금 인상이 고용에 미치는 인과적인 효과

이상적인 인과적 효과

= (뉴저지에서 최저임금 인상 후 고용률)

– (만약 뉴저지에서 최저임금이 인상되지
않았다면 있었을 잠재적 고용률)

반사실 (counterfactual)

- (자연실험의 기본 아이디어) 관찰할 수 없는 반사
실을 대신할 수 있는 비교가능한 대상을 찾자!

자연실험을 통한 인과적 효과

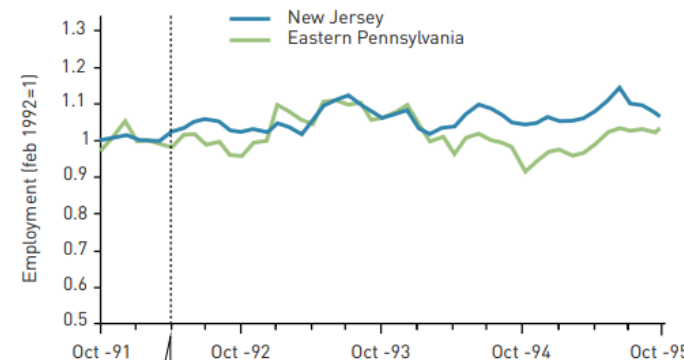
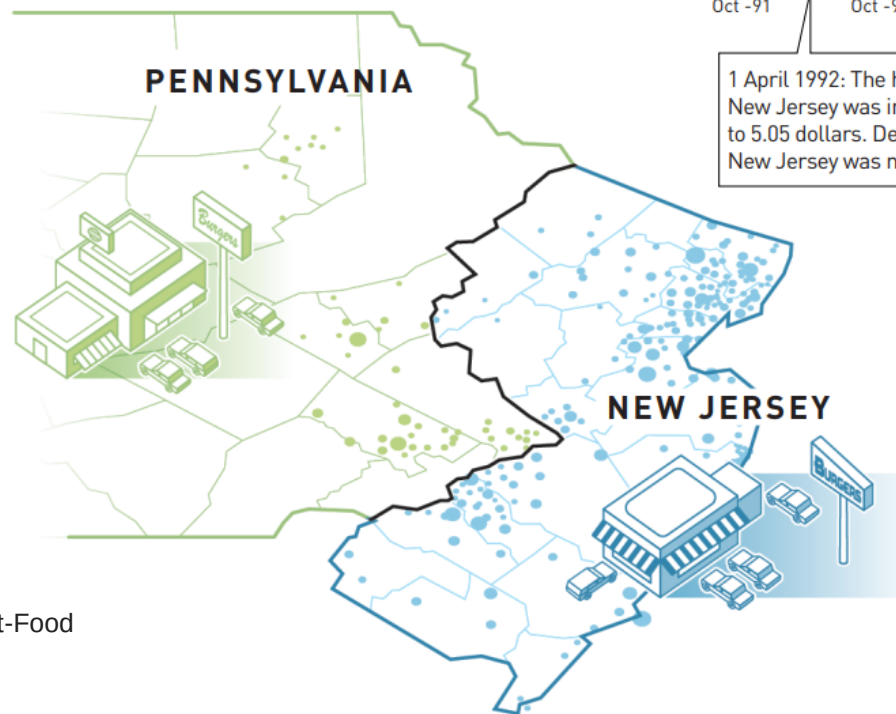
= (뉴저지에서 최저임금 인상 후 고용률)

– (최저임금이 인상되지 않았고 뉴저지와 인접해서
생활환경이 유사한 펜실베이니아에서의 고용률)

Card and Krueger used a natural experiment to study how increasing the minimum wage affects employment.

The researchers identified a treatment group (restaurants in New Jersey) and a control group (restaurants in eastern Pennsylvania) to measure the effect of increasing the minimum wage.

● CONTROL GROUP ● TREATMENT GROUP



1 April 1992: The hourly minimum wage in New Jersey was increased from 4.25 dollars to 5.05 dollars. Despite this, employment in New Jersey was not affected.

Card, D. and Krueger, A.B., 1994. Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. *American Economic Review*, 84(4), pp.772-93.

자연실험 사례 (2) 이민자 유입이 노동시장에 미치는 영향 (Card 1990)

- 1980년 약 12만 여명의 쿠바인들이 피델 카스트로의 사회주의를 피해 탈출한 마리엘 보트리프트(Mariel Boatlift) 사건으로 인해 미국 마이애미에 예상치 못한 대규모의 이민자가 유입되면서 자연실험 상황 연출



이상적인 인과적 효과

= (마이애미에서 마리엘 보트리프트 이후 고용률)

- (만약 마이애미에 쿠바 이민자가 유입되지 않았다면 있었을 잠재적 고용률)

반사실 (counterfactual)

- (자연실험의 기본 아이디어) 관찰할 수 없는 반사실을 대신 할 수 있는 비교가능한 대상을 찾자!

자연실험을 통한 인과적 효과

= (마이애미에서 마리엘 보트리프트 후 고용률)

- (마리엘 보트리프트의 영향이 전혀 없었고, 사회경제적 측면에 있어서 마이애미와 유사한 애틀란타, 휴스턴, 로스앤젤레스, 탬파에서의 고용률)

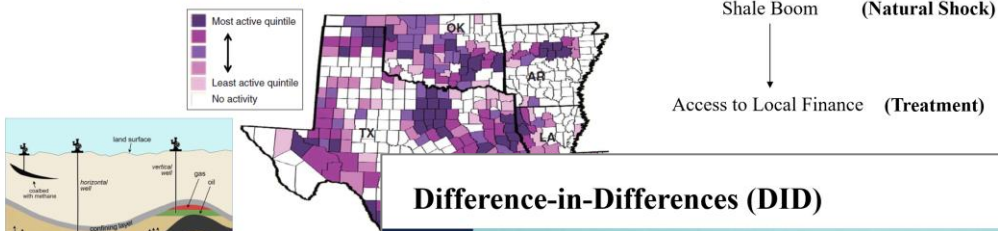
Card, D., 1990. The Impact of the Mariel Boatlift on the Miami Labor Market . *ILR Review*, 43(2), pp.245-257.

[서머세션] 인과추론을 위한 연구 디자인과 분석 도구

Examples of Exogenous Shock for Quasi-Experiments

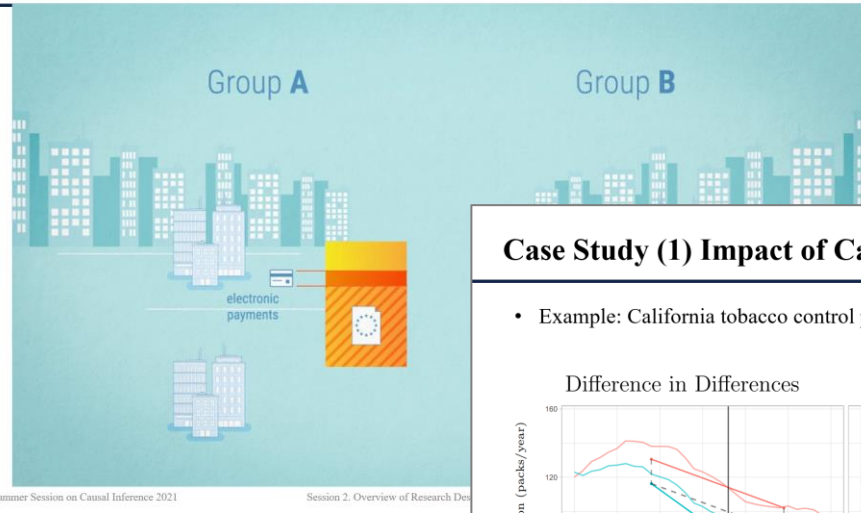
- Example: Impact of the access to local finance on firm formation (Gilje 2019)

Figure 1. (Color online) Location and Intensity of Shale Activity



[Session 2-3] 실험 아닌, 실험 같은 준실험 (Quasi-Experiment)

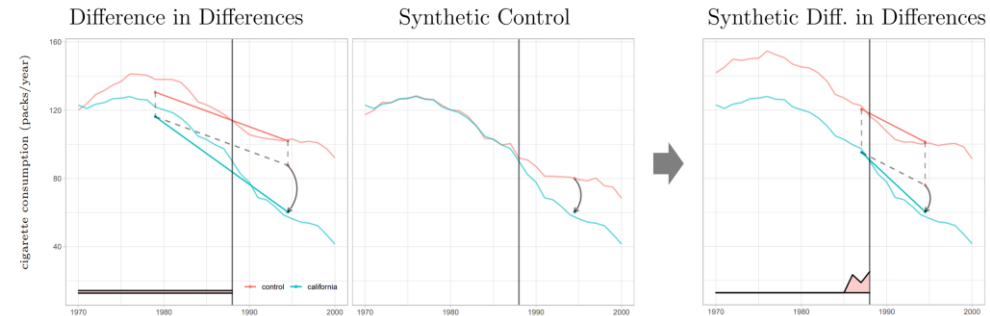
Difference-in-Differences (DID)



[Session 2-4] 준실험 분석도구: 이중차분법 & 회귀불연속 (Difference-in-Differences & Regression Discontinuity)

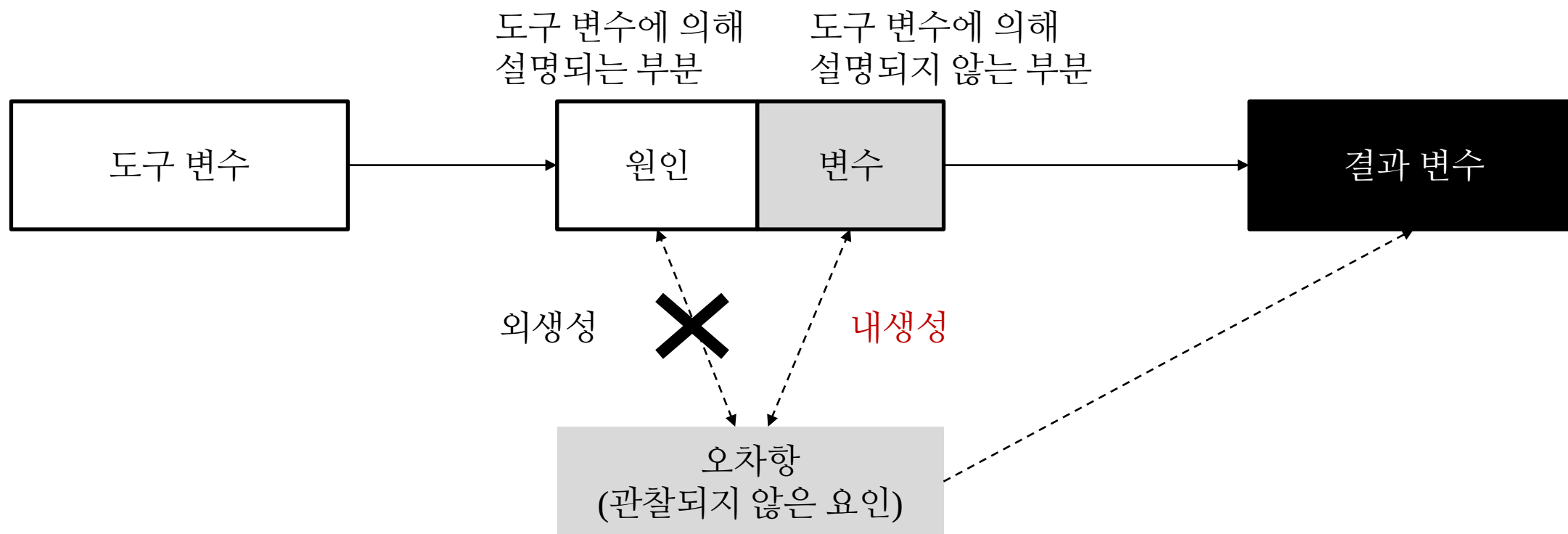
Case Study (1) Impact of California Anti-Tobacco Legislation

- Example: California tobacco control program (Proposition 99) (Abadie et al. 2010)

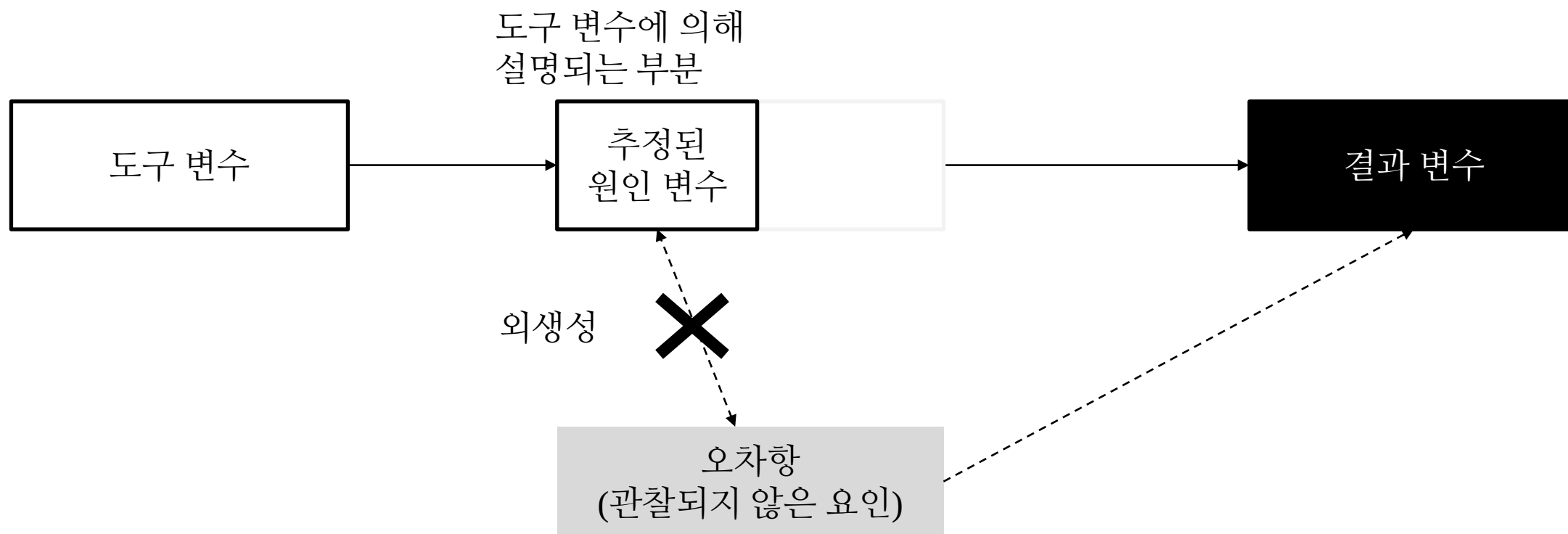


[Session 18-1] 가상의 통제집단 (Synthetic Control)

- 도구 변수 (Instrumental Variable): 원인 변수와 상관성을 갖지만, 원인 변수를 제외하고 결과 변수에 영향을 줄 수 있는 다른 요인들과는 독립적인 변수



- 2-Stage Least Squares (2SLS): 도구 변수를 활용하여 원인 변수 중에서 오차항과 관련되지 않은 외생적인 부분만 추정한 후(1단계 추정), 도구 변수에 의해 추정된 외생적인 원인 변수의 인과적인 효과를 추정(2단계 추정)



도구 변수는 인과 관계를 추론하는 적절한 도구일까?



- 도구 변수의 역사는 매우 오래 되었지만, 그 동안 비편향적인 추정(unbiased estimate)을 위한 계량경제적/통계적 도구로 인식되어 왔음.

Z: 도구변수, X: 원인변수, Y: 결과변수

First Stage: $X = \alpha_0 + \alpha_1 Z + v$

Second Stage: $Y = \beta_0 + \beta_1 X + u$

Reduced Form: $Y = \gamma_0 + \gamma_1 Z + \delta$

Desired Properties	Requirement Assumptions
$\hat{\beta}_{OLS}$ is an unbiased estimate of β	$E[\epsilon_i] = 0$
$\hat{\beta}_{OLS}$ is an unbiased estimate of β $\hat{\beta}_{OLS}$ is the BLUE estimator	$E[\epsilon_i] = 0$ $Var(\epsilon_i) = \sigma^2 < \infty, \forall i$ $Cov(\epsilon_i, \epsilon_j) = 0, \forall i \neq j$
$\hat{\beta}_{OLS}$ is an unbiased estimate of β $\hat{\beta}_{OLS}$ is the BLUE estimator $\hat{\beta}_{OLS}$ is mathematically equivalent to the Maximum Likelihood Estimator $\hat{\beta}_{MLE}$	$E[\epsilon_i] = 0$ $Var(\epsilon_i) = \sigma^2 < \infty, \forall i$ $Cov(\epsilon_i, \epsilon_j) = 0, \forall i \neq j$ $\epsilon_1, \dots, \epsilon_n \sim Normal$, are identically and independently distributed (i.i.d.)

도구 변수는 인과 관계를 추론하는 적절한 도구일까?

- Angrist & Imbens 의 공로: 도구 변수를 인과추론을 위한 잠재적결과 프레임워크로 이해할 수 있는 기틀을 마련함.

Causal inference and potential outcomes

Unit	Treatment status T_i	Outcome under treatment $Y_i(1)$	Outcome under no treatment $Y_i(0)$	Covariates X_i
1	1	✓	?	✓
2	1	✓	?	✓
3	0	?	✓	✓
4	0	?	✓	✓

Z: 도구변수, X: 원인변수, Y: 결과변수

First Stage: $X = \alpha_0 + \alpha_1 Z + v$

Second Stage: $Y = \beta_0 + \beta_1 X + u$

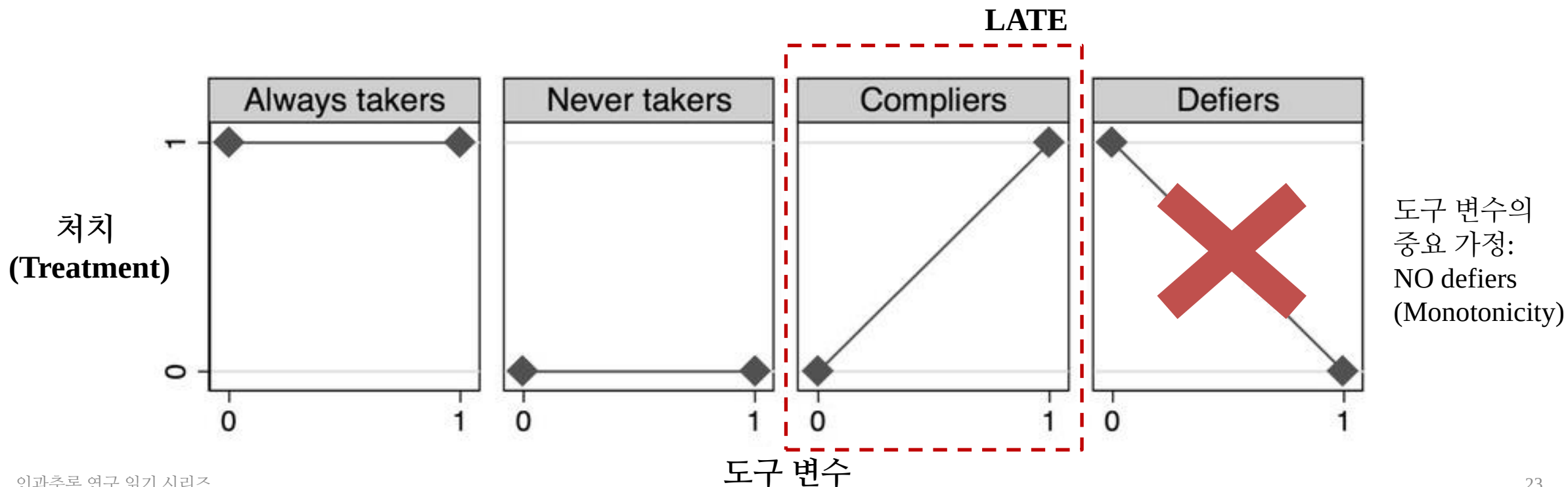
Reduced Form: $Y = \gamma_0 + \gamma_1 Z + \delta$

- 인과추론: 랜덤화 실험 및 준실험/자연실험을 통한
Average Treatment Effect (ATE) 추정

Imbens, G.W. and Angrist, J.D., 1994. Identification and Estimation of Local Average Treatment Effects. *Econometrica*, 62(2), pp.467-475.

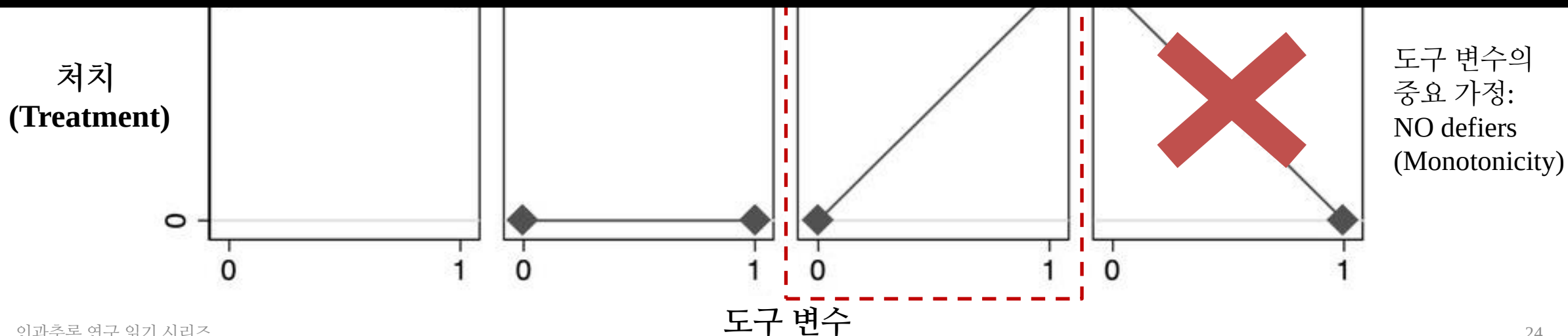
Angrist, J.D., Imbens, G.W. and Rubin, D.B., 1996. Identification of Causal Effects Using Instrumental Variables. *Journal of the American Statistical Association*, 91(434), pp.444-455.

- 잠재적결과 프레임워크는 도구 변수가 인과적 효과로 해석되기 위한 조건을 명확하게 함.
 - (i) Stable Unit Treatment Value Assumption, (ii) Relevance, (iii) Ignobility/Exogeneity, (iv) Exclusion Restriction, (v) Monotonicity
- 도구 변수에 의한 처치 배정(Treatment Assignment) 메커니즘을 체계화 함.
- 도구 변수 추정치는 전체 집단에 대한 Average Treatment Effect (ATE)가 아닌, 도구 변수에 의해 유도되는 처치에 영향을 받는 사람들(Compliers)에 대한 효과인 **Local Average Treatment Effect (LATE)**를 추정한다는 것을 밝힘.



- 잠재적결과 프레임워크는 도구 변수가 인과적 효과로 해석되기 위한 조건을 명확하게 함.
 - (i) Stable Unit Treatment Value Assumption, (ii) Relevance, (iii) Ignobility/Exogeneity, (iv) Exclusion Restriction, (v) Monotonicity
- 도구 변수에 의한 처치 배정(Treatment Assignment) 메커니즘을 체계화 함.
- 도구 변수 추정치는 전체 집단에 대한 Average Treatment Effect (ATE)가 아닌, 도구 변수에 의해 유도되는 처치에

도구 변수는 처치를 유도하는 또다른 (가상의) 준실험/자연실험이다.



[Appendix] 도구 변수와 Local Average Treatment Effect



$$Z \rightarrow D(Z) \rightarrow Y(Z, D)$$

Z: 도구변수 (0 or 1), D: 원인변수 (0 or 1), Y: 결과변수

잠재적결과
프레임워크

$$\begin{aligned} & E[Y_i(Z = 1, D_i(Z = 1)) - Y_i(Z = 0, D_i(Z = 0))] \\ &= E[Y_i(D_i(1)) - Y_i(D_i(0))] \end{aligned}$$



Stable Unit Treatment Value Assumption & Exclusion Restriction

$$= E\left[[Y_i(1)D_i(1) + Y_i(0)(1 - D_i(1))] - [Y_i(1)D_i(0) + Y_i(0)(1 - D_i(0))]\right]$$

$$= E[(Y_i(1) - Y_i(0))(D_i(1) - D_i(0))]$$

$$\begin{aligned} &= E[Y_i(1) - Y_i(0) \mid D_i(1) - D_i(0) = 1]P[D_i(1) - D_i(0) = 1] \\ &\quad - E[Y_i(1) - Y_i(0) \mid D_i(1) - D_i(0) = -1]P[D_i(1) - D_i(0) = -1] \end{aligned}$$

$$= E[Y_i(1) - Y_i(0) \mid D_i(1) - D_i(0) = 1]P[D_i(1) - D_i(0) = 1]$$

$$= E[Y_i(1) - Y_i(0) \mid D_i(1) - D_i(0) = 1]E[D_i(1) - D_i(0)]$$

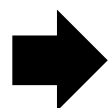


Monotonicity



Monotonicity & Relevance

$$\begin{aligned} & E[D_i(1) - D_i(0)] \\ &= 1 \times P[D_i(1) - D_i(0) = 1] + 0 \times P[D_i(1) - D_i(0) = 0] \end{aligned}$$



$$\frac{E[Y_i(Z = 1, D_i(Z = 1)) - Y_i(Z = 0, D_i(Z = 0))]}{E[D_i(1) - D_i(0)]} = E[Y_i(1) - Y_i(0) \mid D_i(1) - D_i(0) = 1]$$



Compliers

IV Estimand

Local Average Treatment Effect

도구 변수 사례 (1) 의무 교육이 소득에 미치는 영향 (Angrist and Krueger 1991)

- 특정 나이까지 의무 교육을 받아야 하는 규정으로 인해 태어난 분기에 따라 학습 기간의 차이가 발생
- 태어난 분기를 도구 변수로 활용하여 의무 교육 규정 때문에 교육 연수가 달라질 수 있는 사람들(compliers)에 대해서 의무교육이 소득에 미치는 인과적인 효과 추정

Figure 2: Years of schooling by quarter of birth

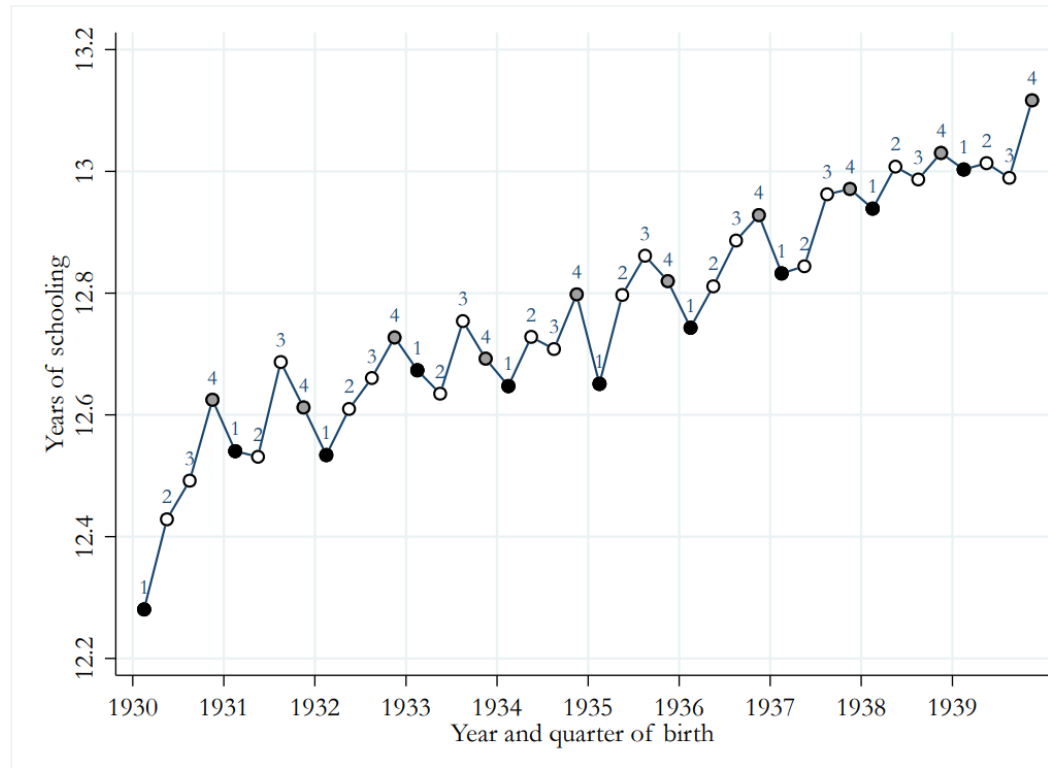
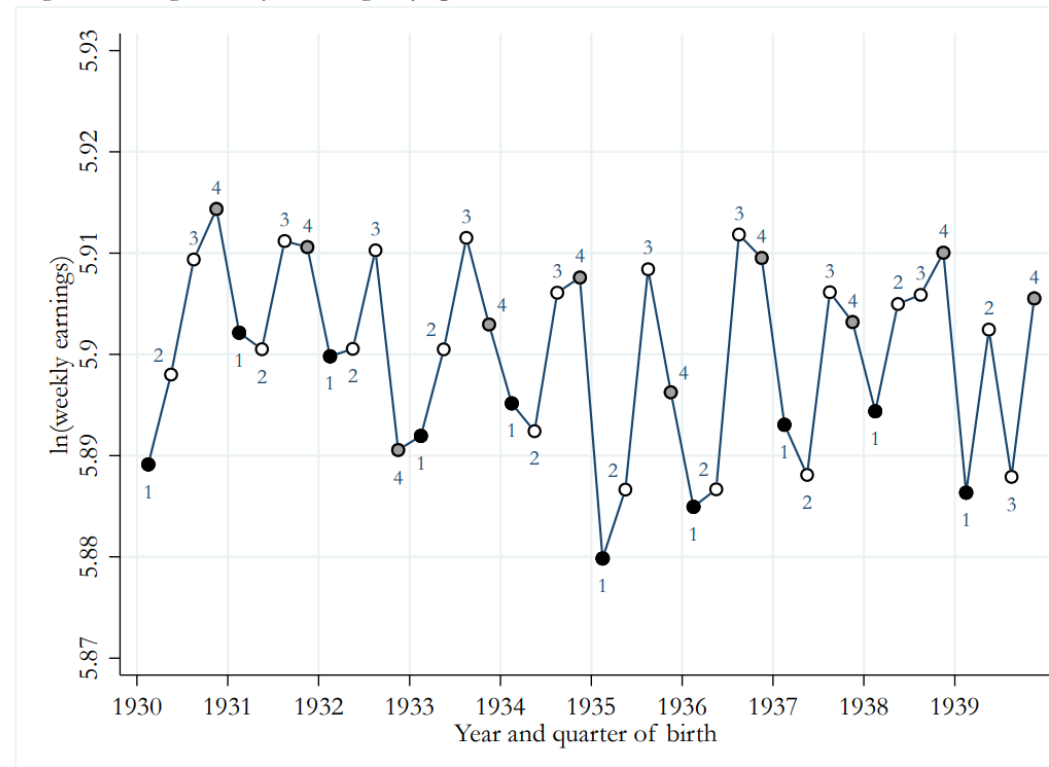


Figure 3: Log weekly earnings by quarter of birth



도구 변수 사례 (1) 의무 교육이 소득에 미치는 영향 (Angrist and Krueger 1991)

- 특정 나이까지 의무 교육을 받아야 하는 규정으로 인해 태어난 분기에 따라 학습 기간의 차이가 발생
- 태어난 분기를 도구 변수로 활용하여 의무 교육 규정 때문에 교육 연수가 달라질 수 있는 사람들(compliers)에 대해서 의무교육이 소득에 미치는 인과적인 효과 추정

Figure 2: Years of schooling by quarter of birth

Figure 3: Log weekly earnings by quarter of birth

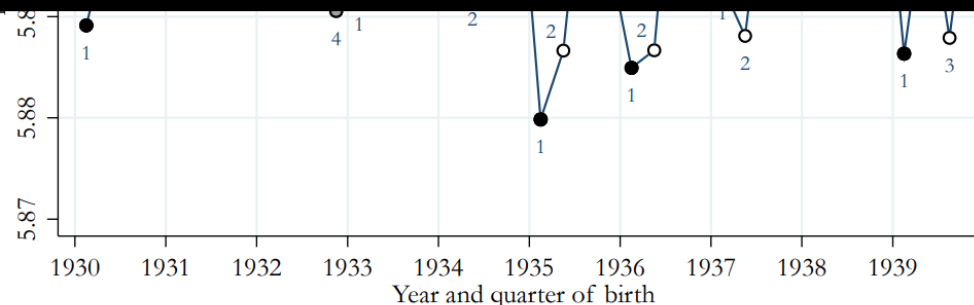
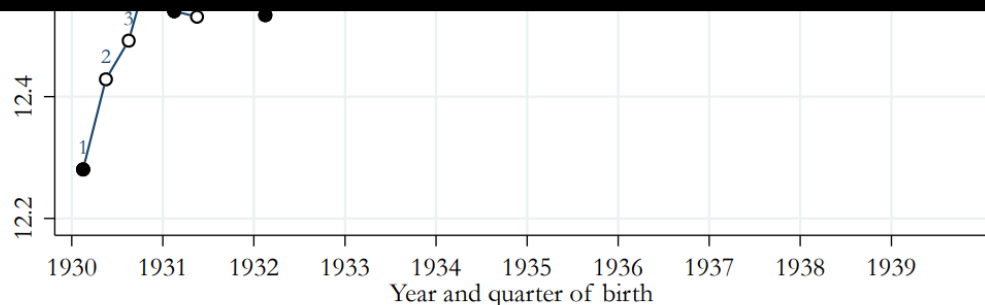
도구 변수 추정 결과 (Local Average Treatment Effect)

≠ 태어난 분기가 소득에 미치는 영향

≠ (모든 사람들에게 대해) 교육 연수가 소득에 미치는 인과적인 효과

≠ (모든 사람들에게 대해) 의무 교육이 소득에 미치는 인과적인 효과

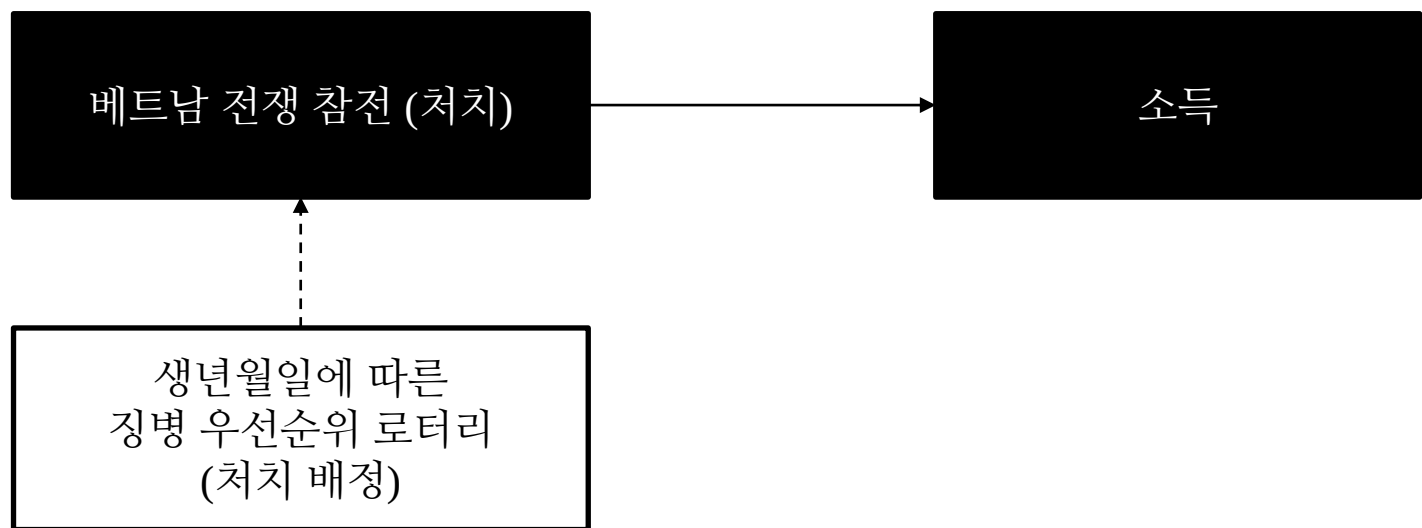
= 의무 교육 규정에 의해 교육 연수가 달라질 수 있는 사람들에게 대해 의무 교육이 소득에 미치는 인과적인 효과



도구 변수 사례 (2) 참전이 소득에 미치는 영향 (Angrist 1990)



- 도구 변수는 랜덤화 실험에서의 불응(noncompliance) 문제를 해결할 수 있음.



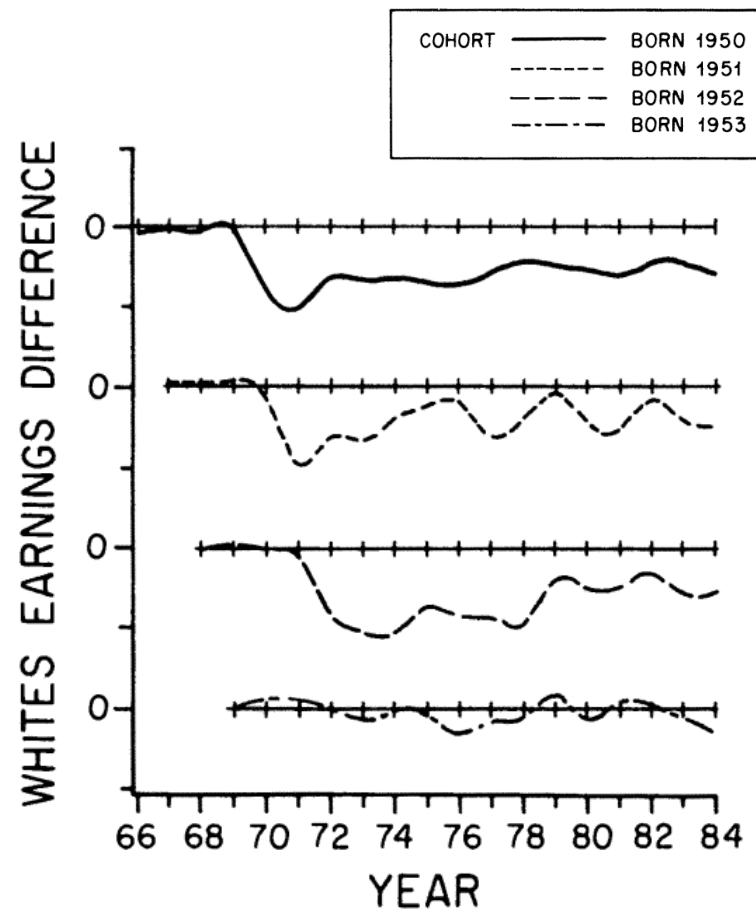
도구 변수 추정 결과 (Local Average Treatment Effect)

≠ 징병 로터리가 소득에 미치는 인과적인 효과 (Intention-to-Treat Effect)

≠ (모든 남성들에 대해) 베트남 참전이 소득에 미치는 인과적인 효과

= 로터리에 의한 징병에 순응하는 남성들에 대해 베트남 참전이 소득에 미치는 인과적인 효과

징병 우선순위 높은 그룹과 낮은 그룹 간의 소득 차이

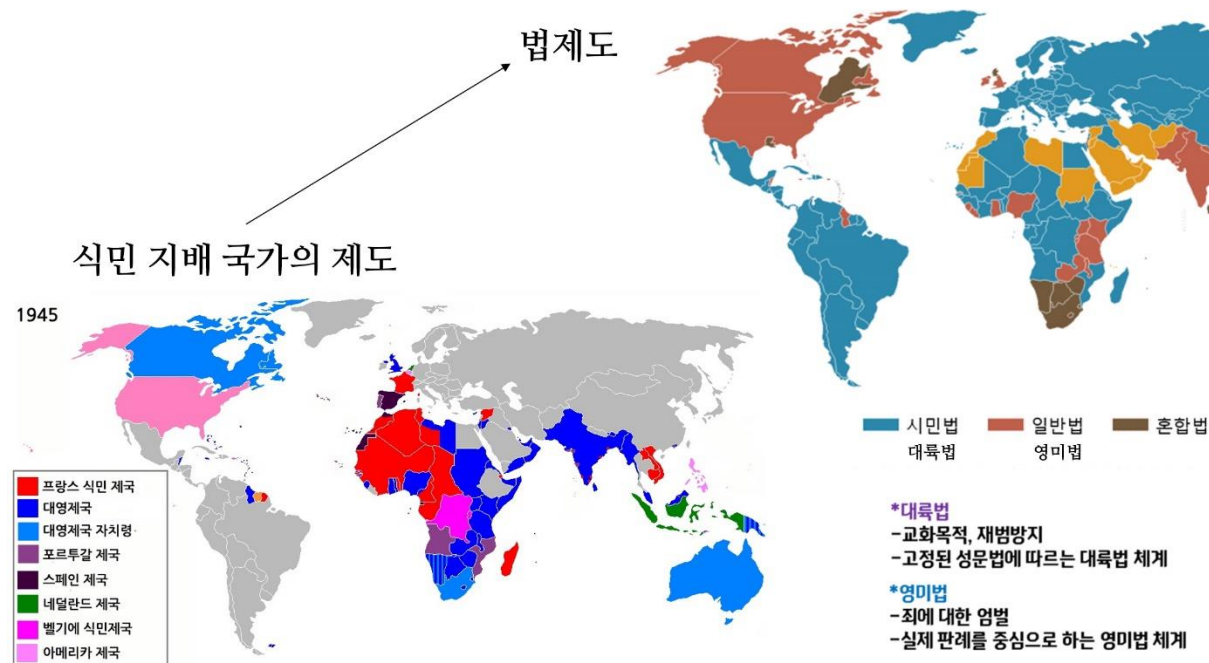


도구 변수 사례 (3) 사회 제도가 경제 성장에 미치는 영향 (Acemoglu and Johnson 2005)

- (자연 실험) 미국과 멕시코 경계에 있는 쌍둥이 도시 간의 비교
- (도구 변수) 유럽국가들의 제국주의 시절, 침략국의 식민지에서의 사망률 → 현재의 사유재산 제도
- (도구 변수) 영국 침략 vs. 대륙국가 (프랑스, 네덜란드 등) 침략 → 현재의 법 제도 및 계약 제도



[인과추론 연구읽기] 남북한 경제 격차의 원인?
사회제도가 경제성장에 미치는 영향 (1) 자연 실험



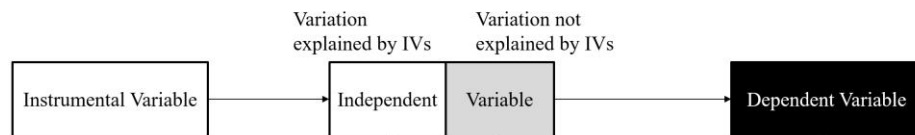
[인과추론 연구읽기] 남북한 경제 격차의 원인?
사회제도가 경제성장에 미치는 영향 (2) 도구 변수

Taking the Selection Bias Out: Instrumental Variable (IV)

What if...? 인과추론의 데이터과학

- Instrumental variables are an instrument to separate the exogenous portion of the independent variable from the endogenous portion (selection bias) out.

[Session 9-1] 도구 변수 (Instrumental Variable)



IV Example (1) Institution and Economic Growth

What if...? 인과추론의 데이터과학

Exogenous

(U

- Settler Mortality during 1500-1900
- Population Density in 1500

Unobserved Factors

Property Rights Institutions in 1990s

Contracting Institutions in 1990s

English Ex-Colonies (English Legal Origin)

How to Report IV Analyses

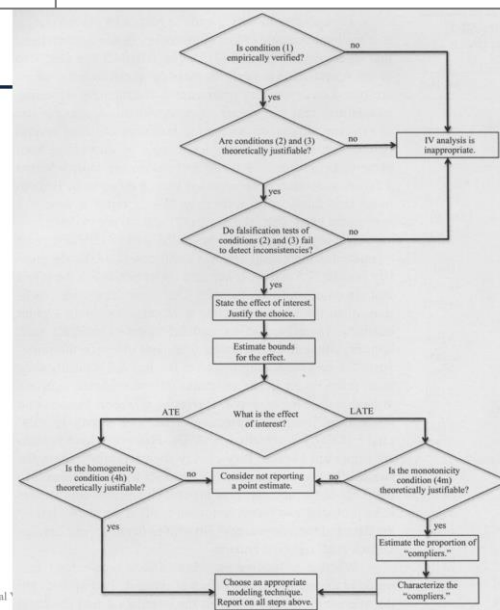
- Step 1. Show statistical tests for the IV validity
- Step 2. Justify theoretical validation of the IVs
- Step 3. Conduct sensitivity tests for the IVs
- Step 4. Clarify whether what you estimate is LATE or ATE.

Swanson, S.A. and Hernán, M.A., 2013. Commentary: how to report instrumental variable analyses (suggestions welcome). *Epidemiology*, 24(3), pp.370-374.

Korea Summer Session on Causal Inference 2021

Session 9. Instrumental

[Session 9-2] 도구 변수의 활용 사례



[Session 9-3] 도구 변수의 활용 팁



Angrist, J.D. and Krueger, A.B., 1991. Does Compulsory School Attendance Affect Schooling and Earnings?. *Quarterly Journal of Economics*, 106(4), pp.979-1014.

Card, D. and Krueger, A.B., 1994. Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. *American Economic Review*, 84(4), pp.772-793.

QUARTERLY JOURNAL OF ECONOMICS

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November 1991

Issue 4

DOES COMPULSORY SCHOOL ATTENDANCE AFFECT SCHOOLING AND EARNINGS?*

JOSHUA D. ANGRIST AND ALAN B. KRUEGER

We establish that season of birth is related to educational attainment because of school start age policy and compulsory school attendance laws. Individuals born in the beginning of the year start school at an older age, and can therefore drop out after completing less schooling. Roughly 25 percent of potential schooling laws. We estimate using quarter of birth as an instrument to estimate the return to education, suggesting that there is little

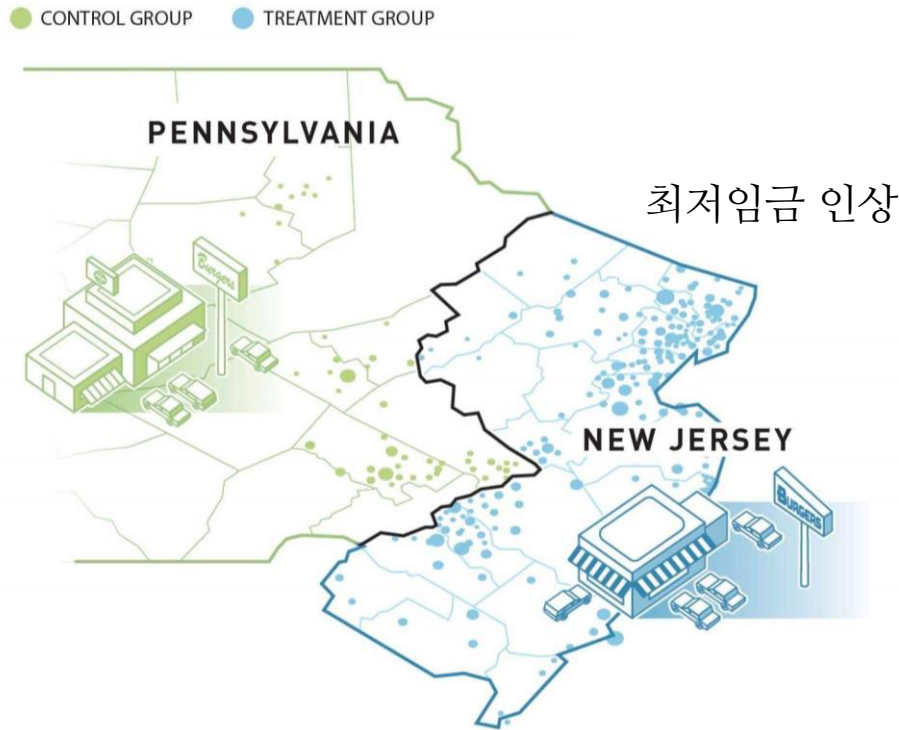
THE AMERICAN ECONOMIC REVIEW

Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania

By DAVID CARD AND ALAN B. KRUEGER*

On April 1, 1992, New Jersey's minimum wage rose from \$4.25 to \$5.05 per hour. To evaluate the impact of the law we surveyed 410 fast-food restaurants in New Jersey and eastern Pennsylvania before and after the rise. Comparisons of employment growth at stores in New Jersey and Pennsylvania (where the minimum wage was constant) provide simple estimates of the effect of the higher minimum wage. We also compare employment changes at stores in New Jersey that were initially paying high wages (above \$5) to the changes at lower-wage stores. We find no indication that the rise in the minimum wage reduced employment. (JEL J30, J23)

- 비판 1. 특정 상황에서의 연구 디자인은 다른 상황에 적용 가능한가? (외부 타당성에 대한 비판)



인과추론 결과에
대한 종합
(Meta Analysis)

1990년대 초반 뉴저지/펜실베이니아에서의 패스트푸드 산업에서의 결과를 우리나라 현실에 그대로 적용할 수 있을까?

Table 2. Evidence of the effects of minimum wage on employment by country group.

Countries	Employment		
	Effect	Number of articles	Percentage
Developed countries	Negative	118	57.84%
	Positive	11	5.39%
	Null	25	12.25 %
	Subtotal	154	75.49%
Developing countries	Negative	36	17.65%
	Positive	5	2.45%
	Null	9	4.41%
	Subtotal	50	24.51%
	Total	204	100%

Source: Own elaboration based on Scopus and Web of Science database.

Martínez, M.J. and Martínez, M.J., 2021. Are the Effects of Minimum Wage on the Labour Market the Same Across Countries? A Meta-Analysis Spanning a Century. *Economic Systems*, 45(1), p.100849.

- 비판 2. 랜덤화실험/준실험/자연실험 연구 디자인이 가능한 주제의 연구에 편중되는 경향에 대한 비판

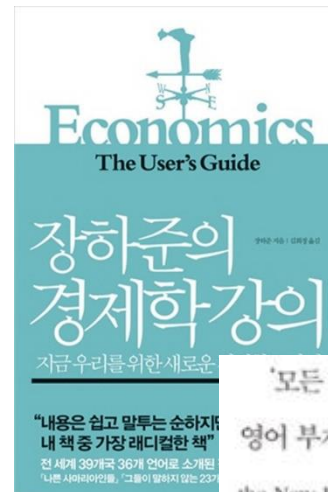
THE NEW REPUBLIC

Noam Scheiber / April 30, 2007

Revenge of the Nerd

A few weeks ago I wrote a piece about how too many academic economists are doing cute and clever work instead of tackling weighty questions ("Freaks and Geeks," April 2). I placed some of the blame for this on Steve Levitt, the University of Chicago professor and author of *Freakonomics*. Levitt, I argued, was both a leading practitioner of cute-and-clever and a role model to top young economists. Now Levitt has responded with a blog post so strange and incoherent it is almost hard to believe he wrote it.

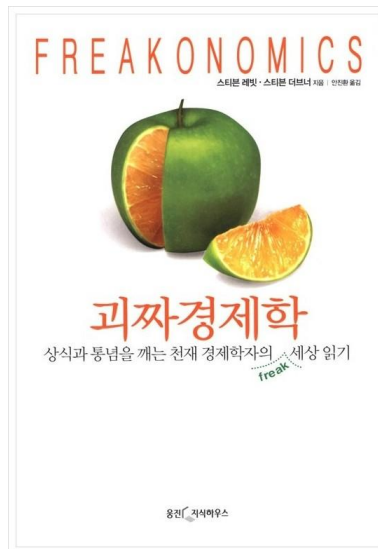
<https://newrepublic.com/article/61521/revenge-the-nerd>



‘모든 것’을 다루는 부분에서도 빠지지 않는다. 『경제학 콘서트2』의 영어 부제는 ‘모든 것에 관한 새로운 경제학의 비밀을 밝힌다(Uncovering the New Economics of Everything)’이다. 현재 전 세계적으로 가장 잘 알려진 경제학 서적 중의 하나인 『괴짜 경제학』의 영어 부제에서 저자 스티븐 레빗과 스티븐 더브너는 경제학이 ‘모든 것의 이면(Hidden Side of Everything)’을 파헤치는 학문이라고 말한다. 로버트 프랭크도 훨씬 겸손하기는 하지만 이와 의견을 같이하는 듯하다. 그의 ‘경제 박물관학’ 시리즈 첫 번째 책의 영어 부제가 ‘왜 경제학은 거의 모든 것을 설명하는가(Why Economics Explains Almost Everything)’ 아닌가?

자, 이 정도면 경제학은 인생, 우주, 그리고 (거의) 모든 것^{??}을 다루는 학문이라 해도 과언이 아닐 것이다.

그러나 생각해 보면 이것은 일반인들이 경제학자들의 본업이라고 여기는 바로 그 분야에서 완전히 실패한 사람들이 주장하기에는 좀 무리가 있는 이야기 아닌가? 바로 경제를 설명하는 일 말이다.



연구 디자인 중심의 인과추론에 대한 비판



인과추론의 데이터과학

- 인과추론 연구가 현실 문제 해결에 어떻게 적용될 수 있는지 궁금한 분들에게 추천하는 하버드 경제학 개론 강의



TRAINING THE NEXT GENERATION

Using Big Data to Solve Economic and Social Problems



A central part of Opportunity Insights' mission is to train the next generation of researchers and policy leaders on methods to study and improve economic opportunity and related social problems. This page provides lecture materials and videos for a course entitled "Using Big Data Solve Economic and Social Problems," taught by Raj Chetty and Greg Bruich at Harvard University.

LECTURE 1
The Geography of Upward Mobility in America

LECTURE 2
Causal Effects of Neighborhoods

LECTURE 3
Moving to Opportunity vs. Place-Based Approaches

LECTURE 4
The American Dream in Historical Perspective

LECTURE 5
Upward Mobility, Innovation, and Economic Growth

LECTURE 6
Higher Education and Upward Mobility

LECTURE 7
The Causal Effect of Colleges

LECTURE 8
Primary Education

LECTURE 9
Teachers and Charter Schools

LECTURE 10
Racial Disparities in Economic Opportunity

LECTURE 11
Improving Health Outcomes

LECTURE 12
The Economics of Health Care and Insurance

LECTURE 13
Improving Judicial Decisions

LECTURE 14
Effects of Air and Water Pollution

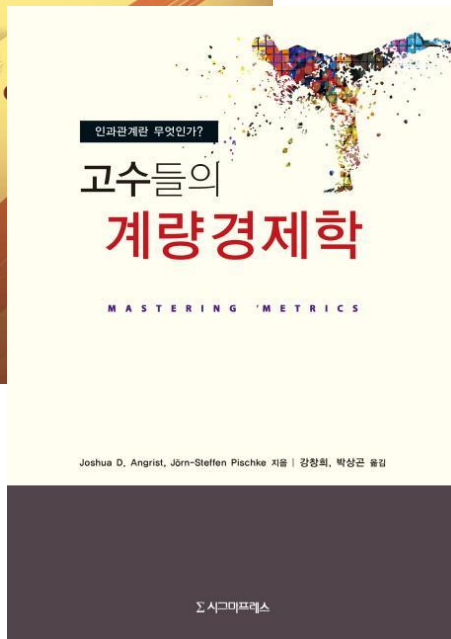
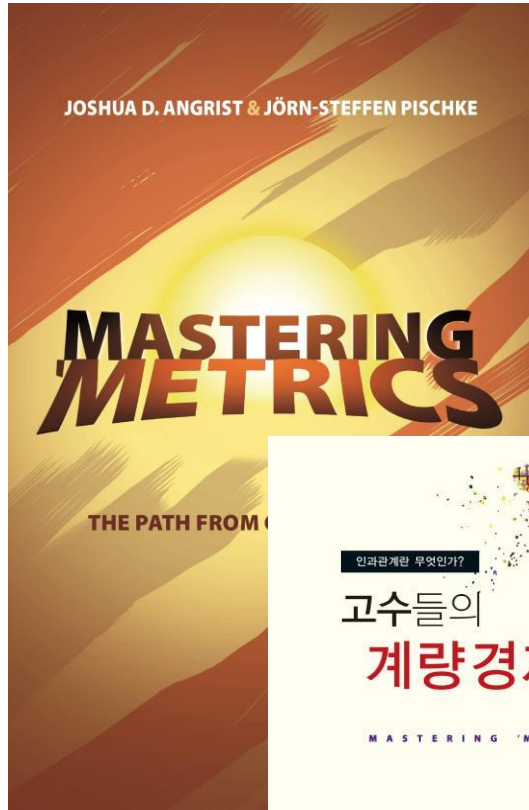
LECTURE 15
Policies to Mitigate Climate Change

LECTURE 16
Income Taxation

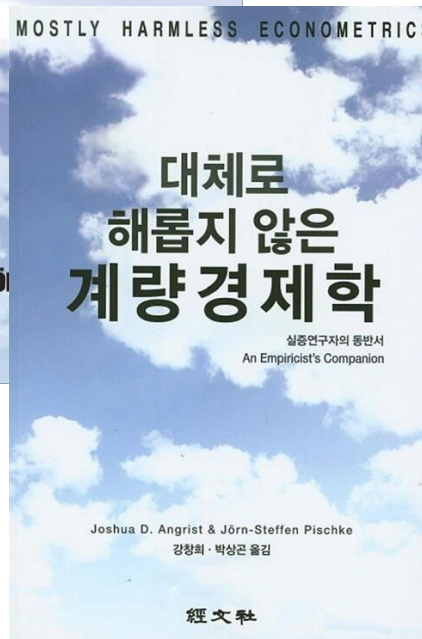
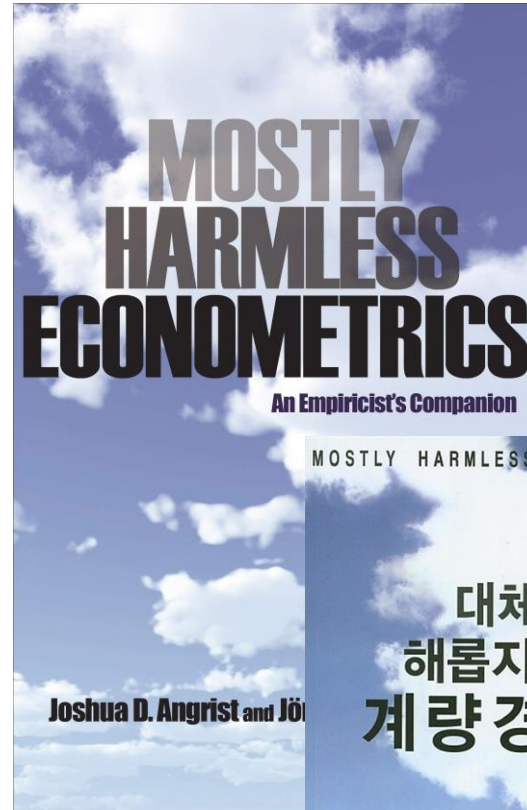
LECTURE 17
Behavioral Public Economics

LECTURE 18
Institutions and Economic Development

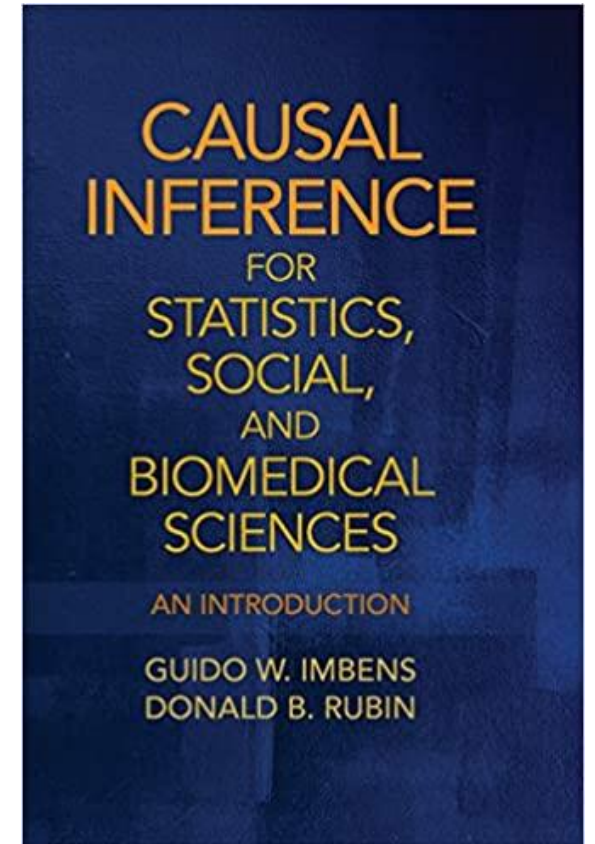
초급



초중급



중급





서머세션은 인과추론 연구방법론과 데이터분석 방법론에 대한 이해를 돕고, 인과추론에 대한 저변을 확대하기 위해 기획된 온라인 세미나입니다. 올해로 5년차를 맞이하는 서머세션은 작년에 이어 올해도 온라인으로 진행하되, 그 범위를 보다 확대하여 계량경제학/통계학 기반의 사회과학/의학 분야 연구 뿐 아니라, 머신러닝을 활용한 인과추론에 대한 주제도 포함할 예정이며, 학계 뿐 아니라 산업계에서의 인과추론의 활용에 대해서도 함께 다루고자 합니다.

The **Korea Summer Session on Causal Inference** aims at introducing research methodologies and data analytical approaches in pursuit of causal inference. Across multiple disciplines, causal inference has been achieved largely building on two frameworks: (i) Potential Outcomes Framework (design-based approach) and (ii) Structural Causal Model (structure-based approach). In addition, machine learning has been rapidly employed for causal inference in ways to improve the quality of causal analysis, estimate causal effects on top of underlying causal structures, and/or discover causal relationships from large quantities of data.

Against this backdrop, the two modules of the summer session are geared toward two learning goals (see [Lecture Materials](#) for details):

[Module 1] To understand research designs for causal inference

[Module 2] To understand how machine learning complements and innovates causal inference

<https://sites.google.com/view/causal-inference2021>

강의 유튜브 채널



인과추론의 데이터과학

데이터 분석을 통해 원인과 결과를 탐구하는 인과추론 (causal inference) 연구를 다룹니다.